Game theoretical inference of human behavior in social networks

Autonomy talk,
ETH Zürich, 21.09.2020

Pagan, N. & Dörfler, F. *Game theoretical inference of human behavior in social networks*. Nature Communications, 2019.

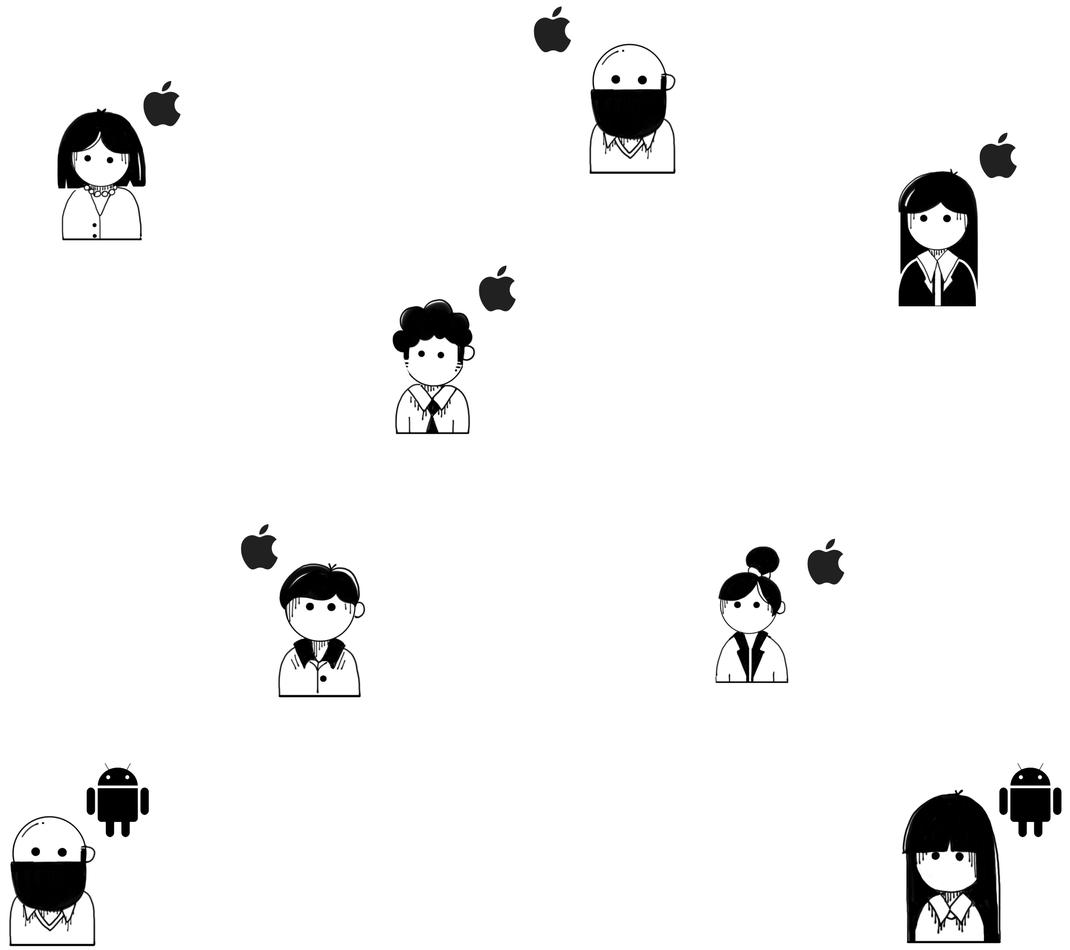
NICOLÒ PAGAN
FLORIAN DÖRFLER

INSTITUT FÜR AUTOMATIK IfA

ETH zürich

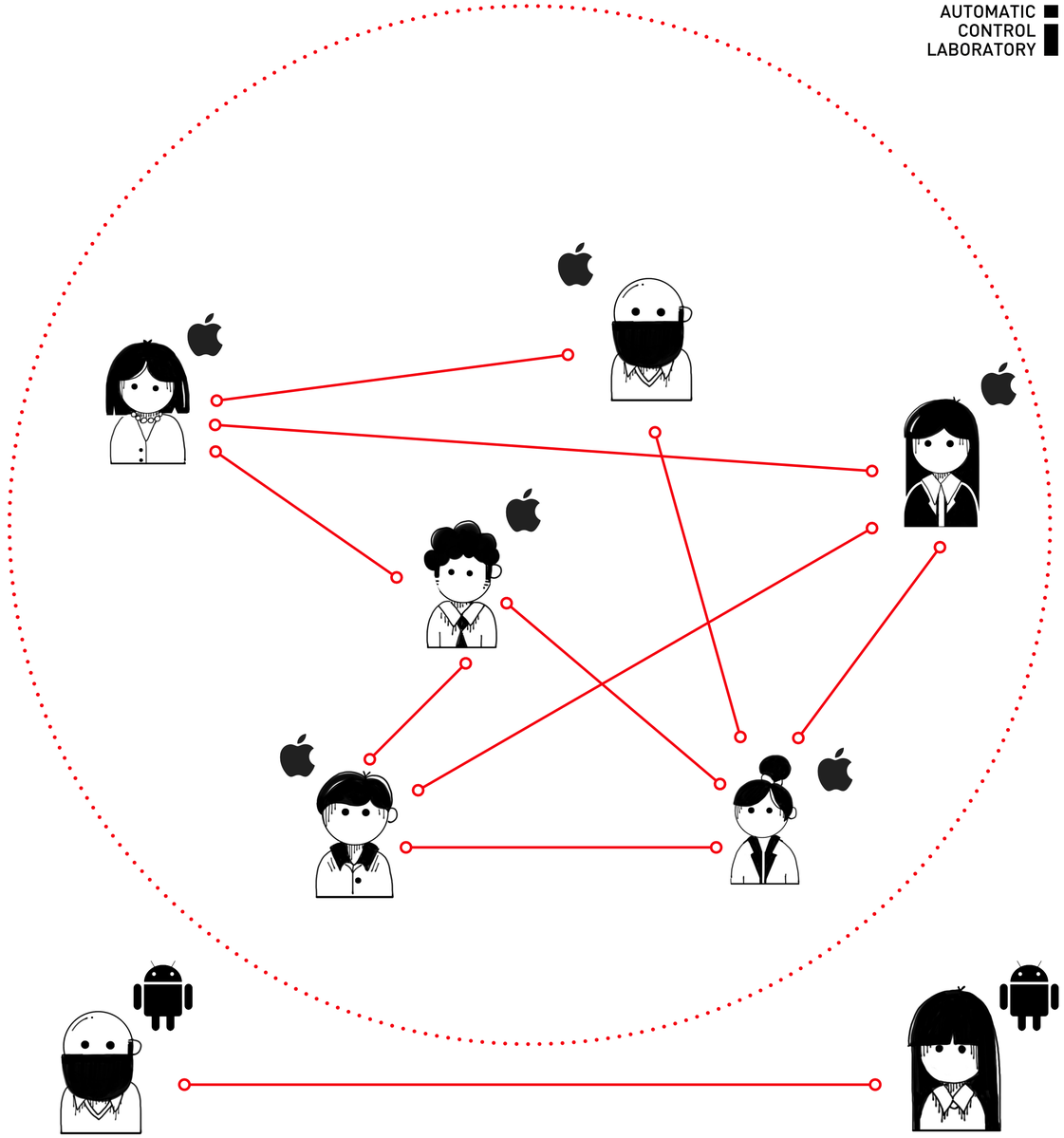
MOTIVATION

How do social networks form?



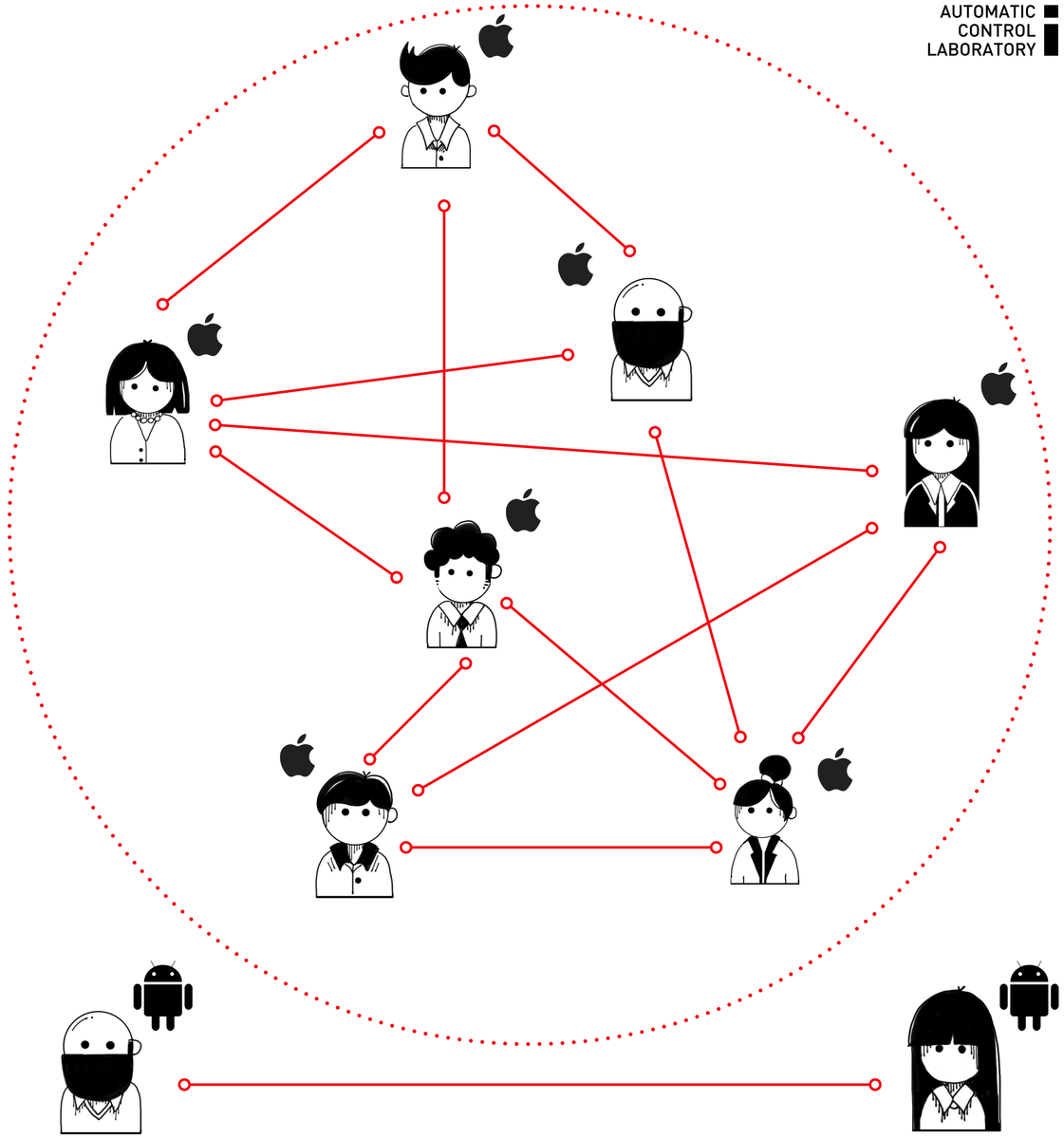
MOTIVATION

- How do social networks form?
- Individual behavior determines the social network structure.



MOTIVATION

- How do social networks form?
- Individual behavior determines the social network structure.
- The social network structure influences individual behavior.



OBSERVATIONS

- Actors decide with whom they want to interact.
  Directionality: *followers* ≠ *followees*

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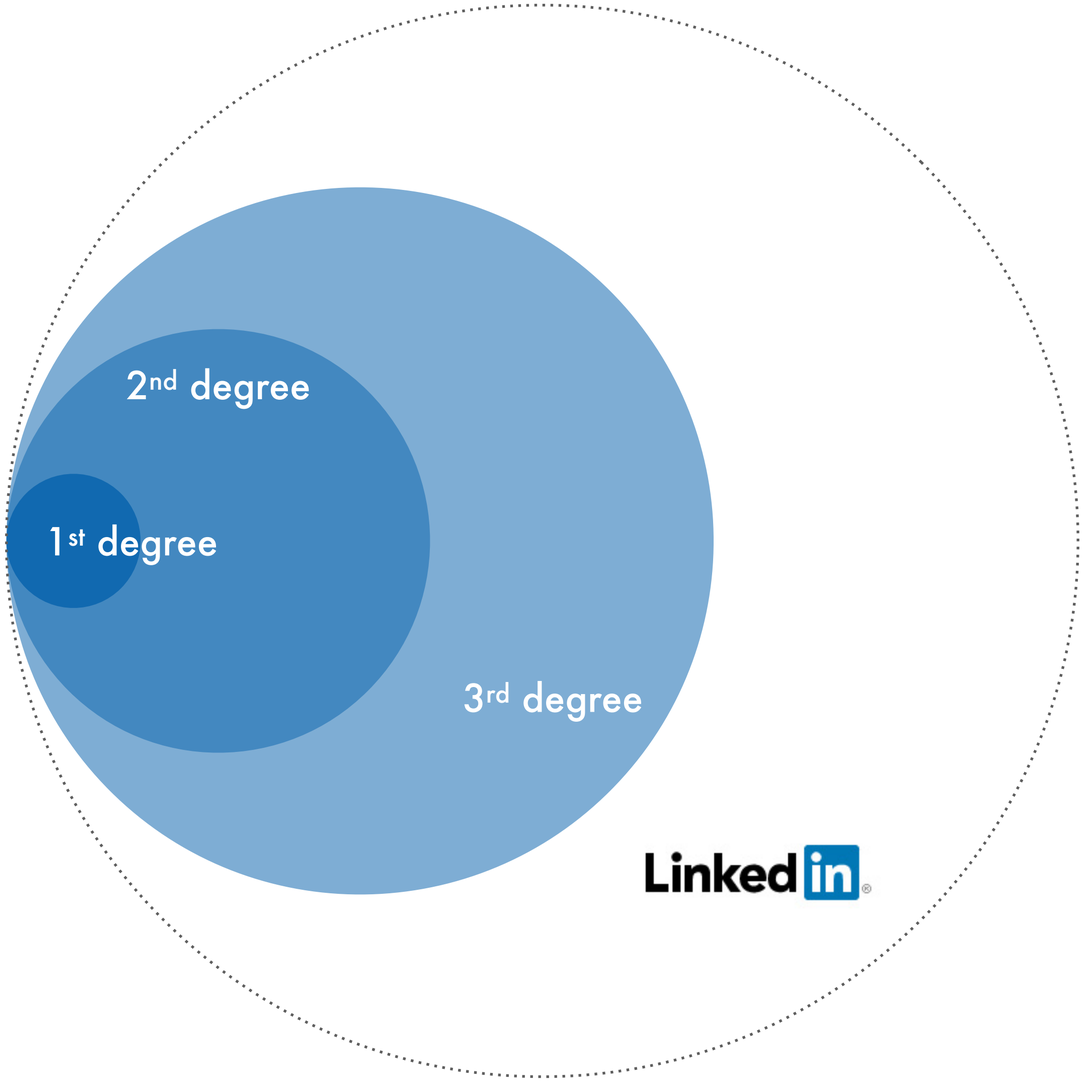
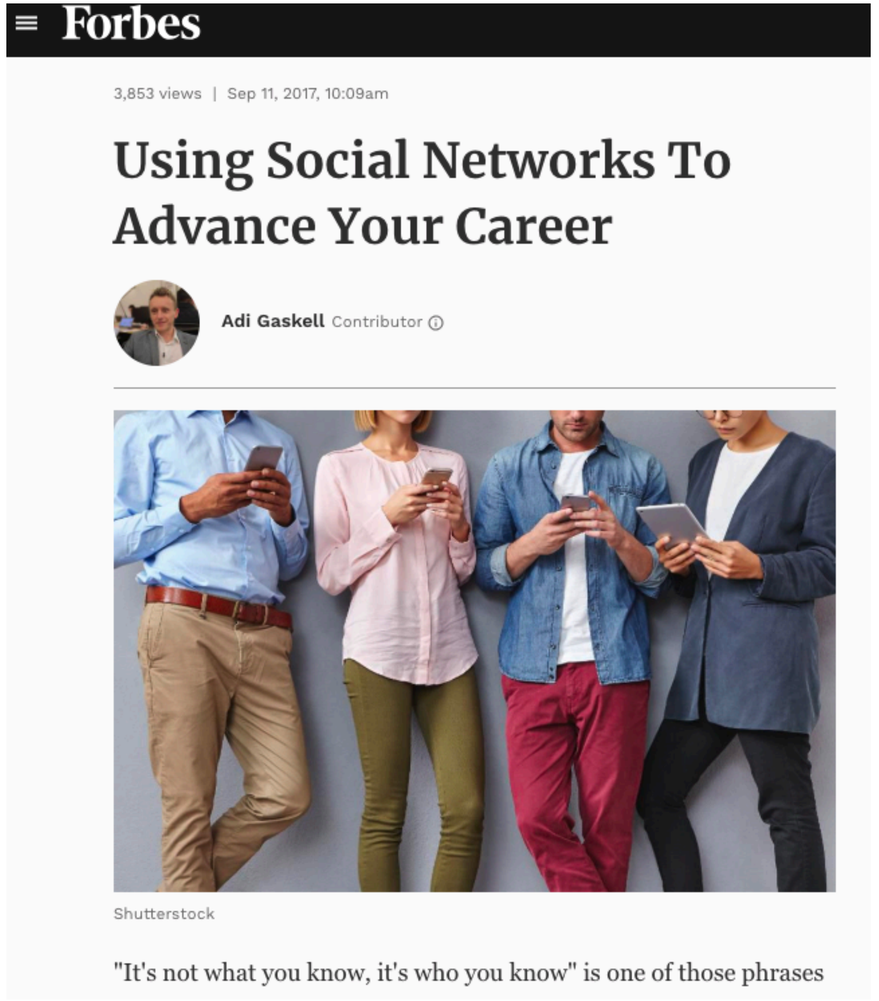


OBSERVATIONS

- Actors decide with whom they want to interact.
  Directionality: *followers ≠ followees*
- Ties can have different **weights**.
- **Limited** information is available.
- Network positions provide **benefits** to the actors.



Forbes

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Using Social Networks To Advance Your Career

Adi Gaskell Contributor

Shutterstock

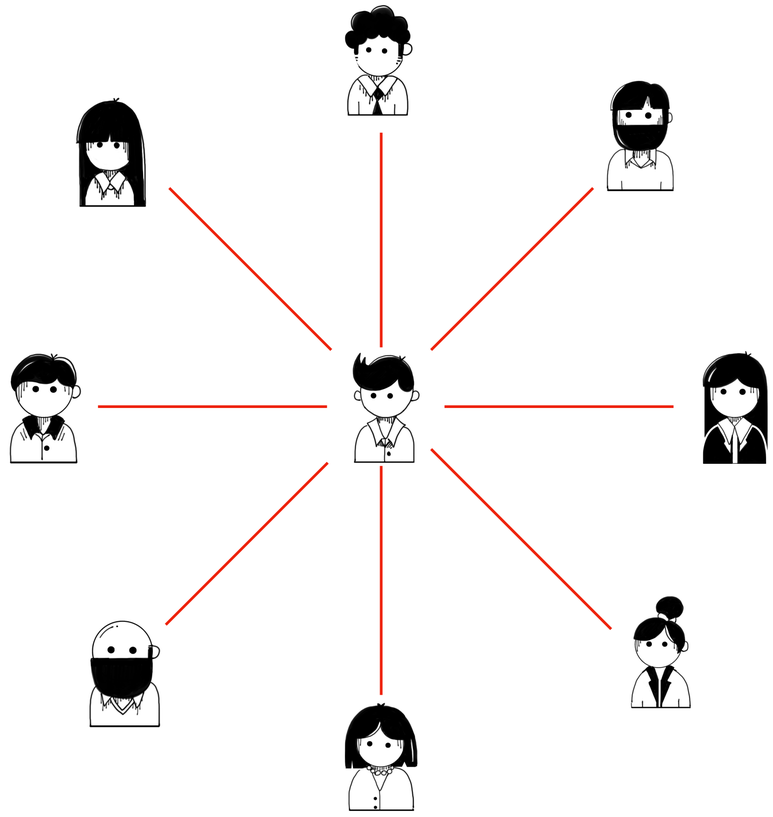
"It's not what you know, it's who you know" is one of those phrases

SOCIAL NETWORK POSITIONS' BENEFITS

Social Influence

The more people we are connected to, the more we can influence them.

[Robins, G. Doing social network research, 2015]



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Social Support

The more our friends' friends are our friends, the safer we feel.

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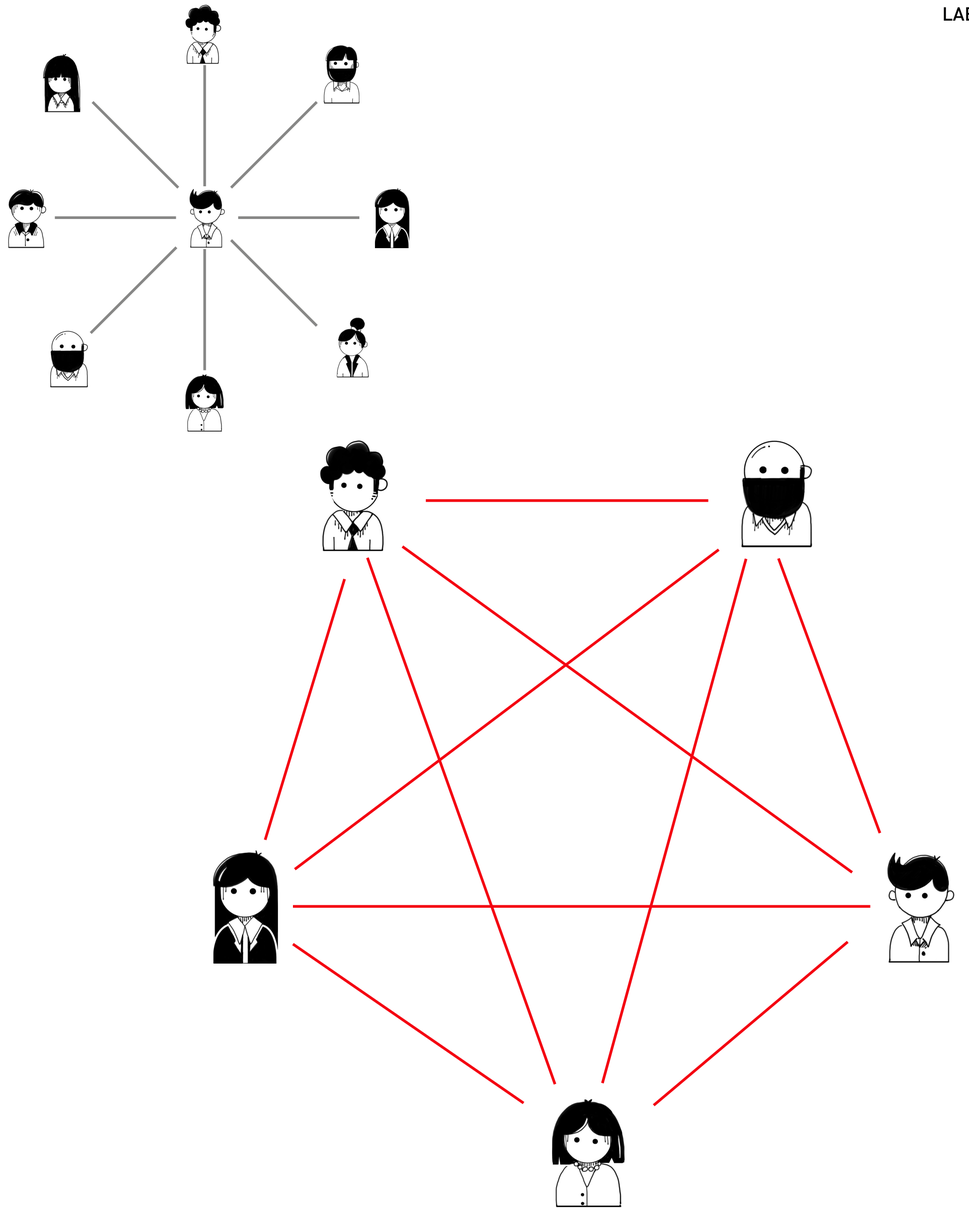

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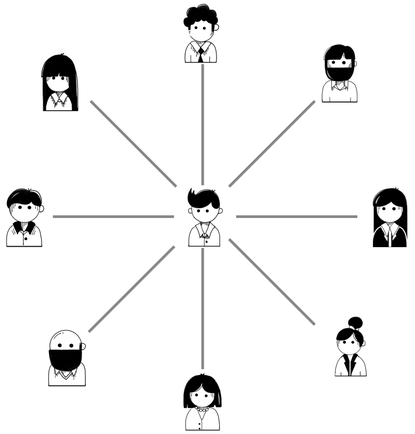

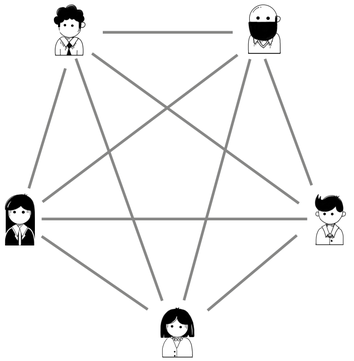

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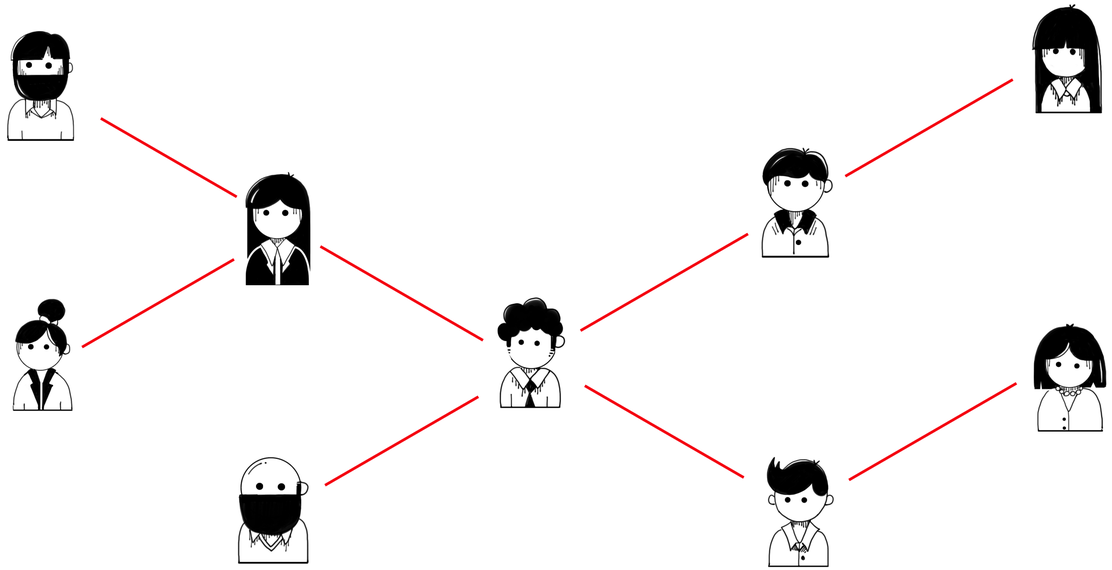

Brokerage

The more we are on the path between people, the more we can control.

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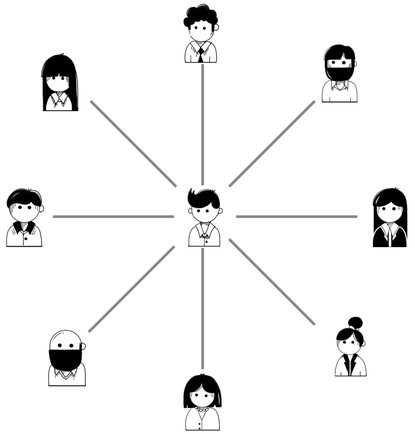
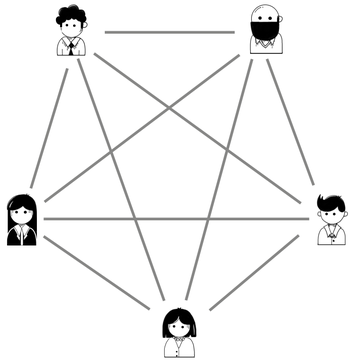
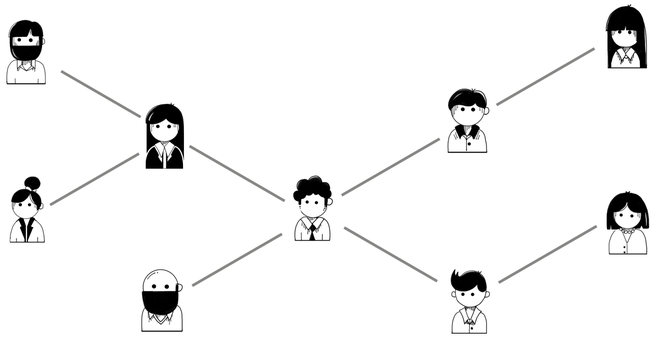


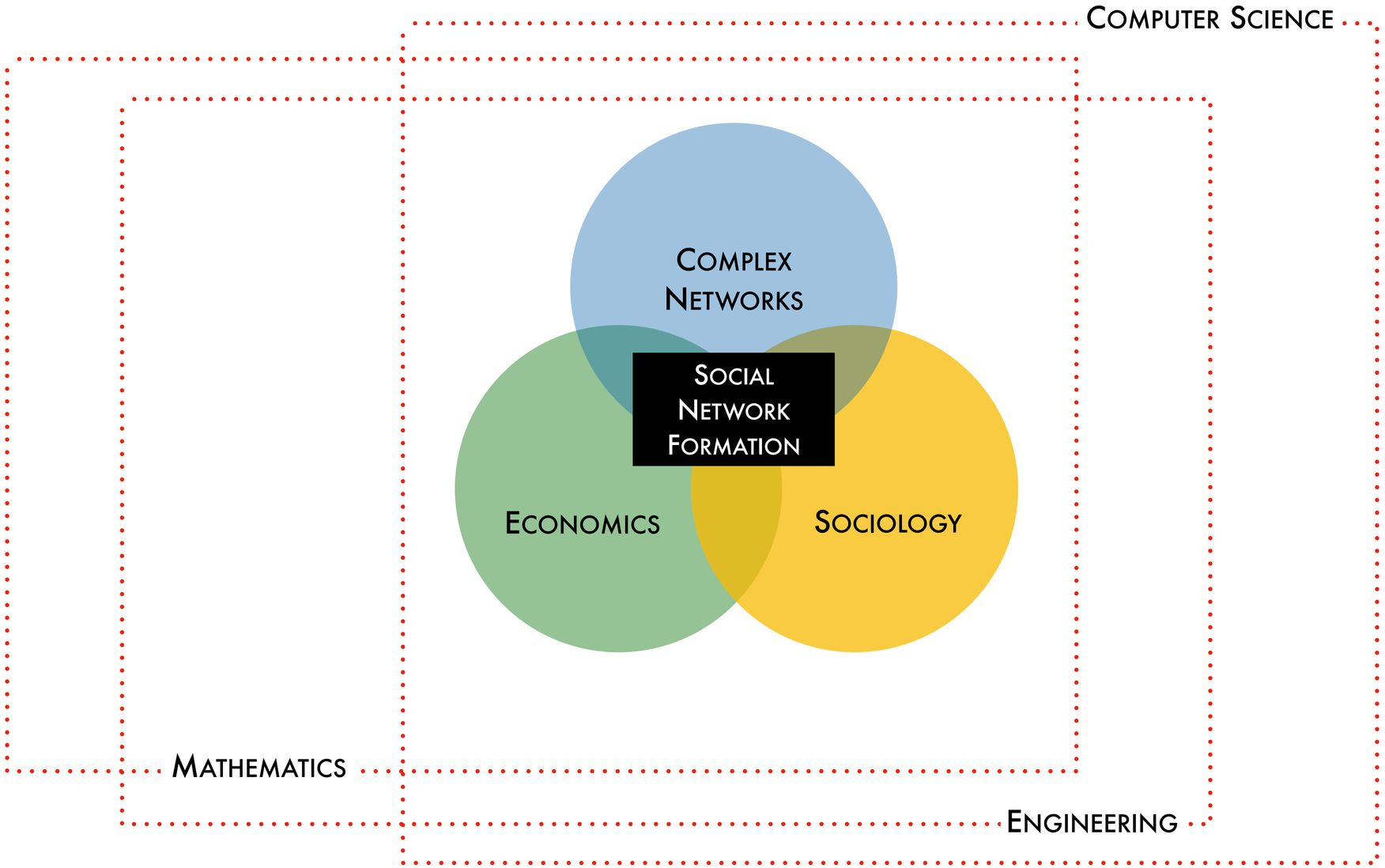
DEGREE CENTRALITY

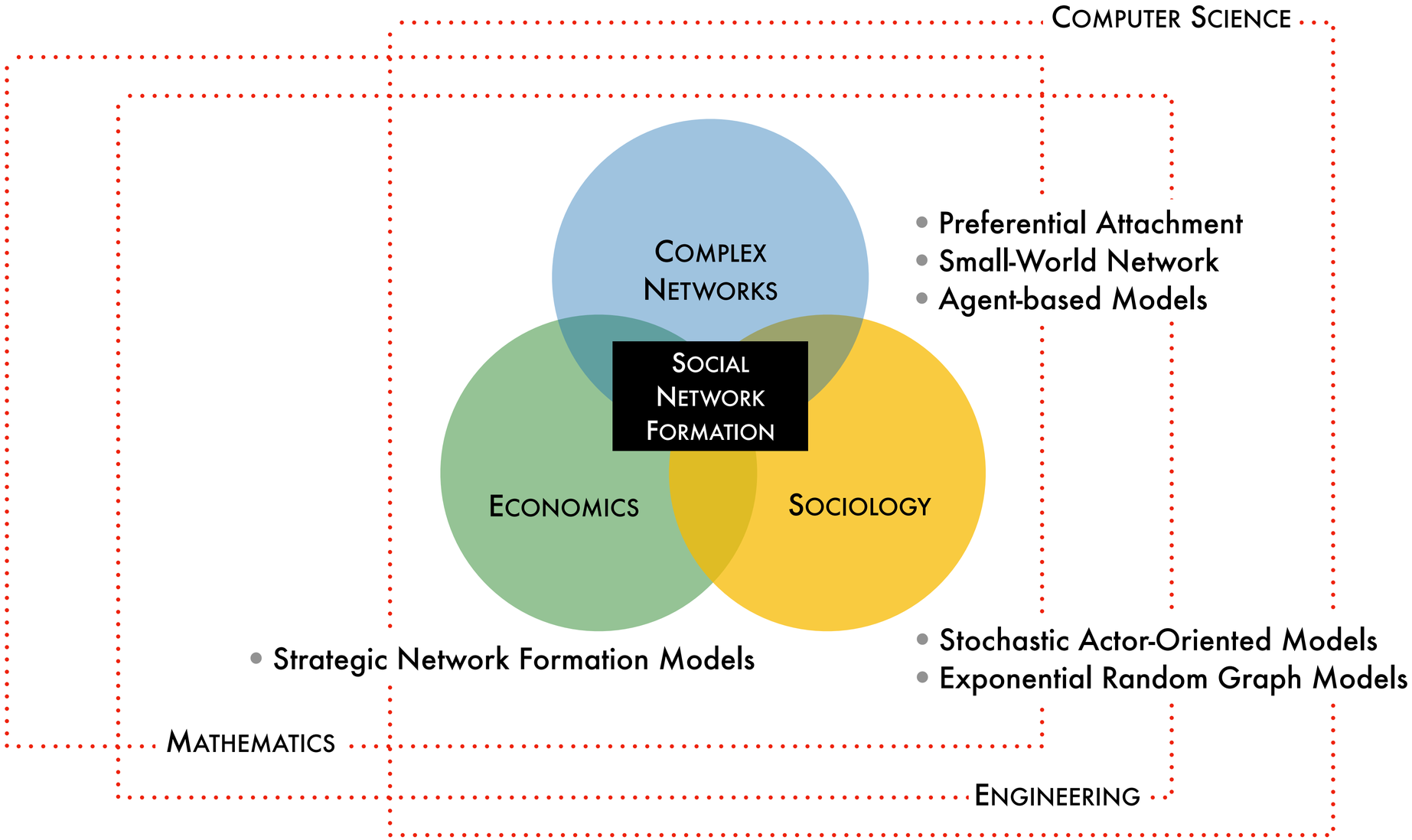


CLUSTERING COEFFICIENT



BETWEENNESS CENTRALITY


COMPUTER SCIENCE
COMPLEX NETWORKS
SOCIAL NETWORK FORMATION
ECONOMICS
SOCIOLOGY
MATHEMATICS
ENGINEERING


COMPUTER SCIENCE
COMPLEX NETWORKS
SOCIAL NETWORK FORMATION
ECONOMICS
SOCIOLOGY
• Preferential Attachment
• Small-World Network
• Agent-based Models
• Strategic Network Formation Models
• Stochastic Actor-Oriented Models
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MATHEMATICS
ENGINEERING

SOCIAL NETWORK FORMATION MODEL

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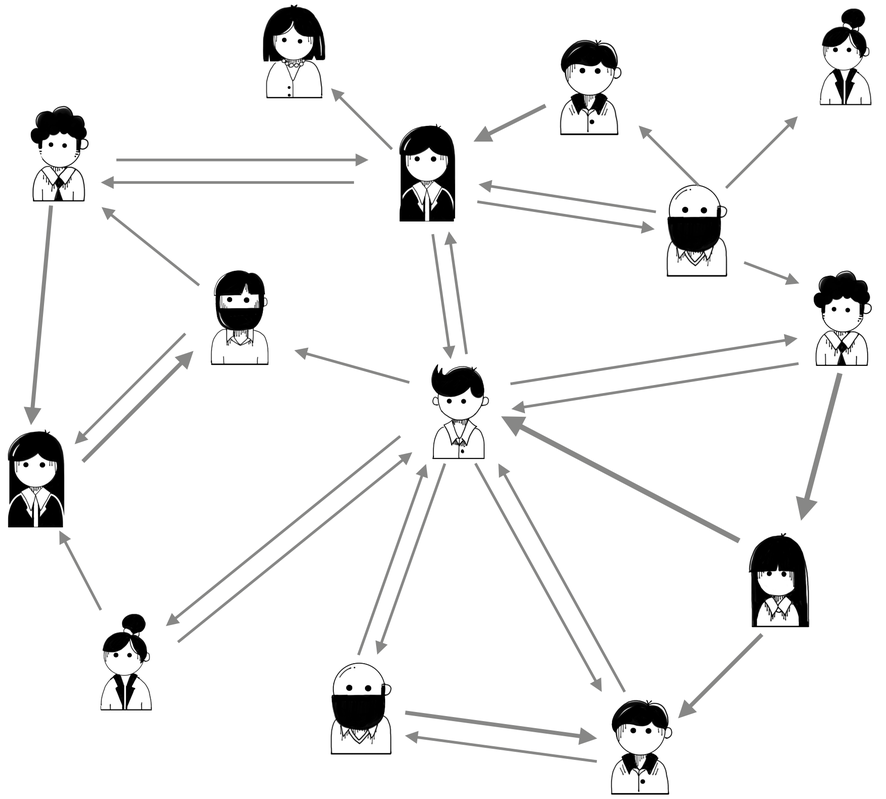
Directed weighted network $\mathcal{G}$ with $\mathcal{N} = \{1,\dots,N\}$ agents. The weight $a_{ij} \in [0,1]$ quantifies the importance of the friendship among i and j from i's point of view.



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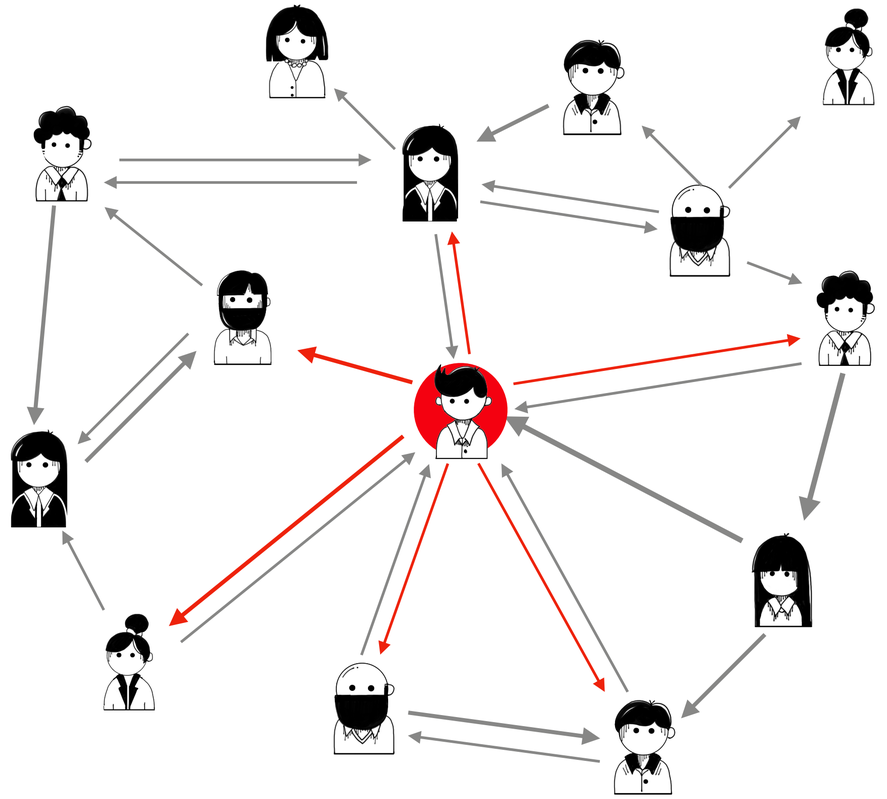
A typical **action** of agent i :

$$a_i = \left[a_{i1}, \dots, a_{i,i-1}, a_{i,i+1}, \dots, a_{iN}\right] \in \mathcal{A} = \left[0,1\right]^{N-1}$$



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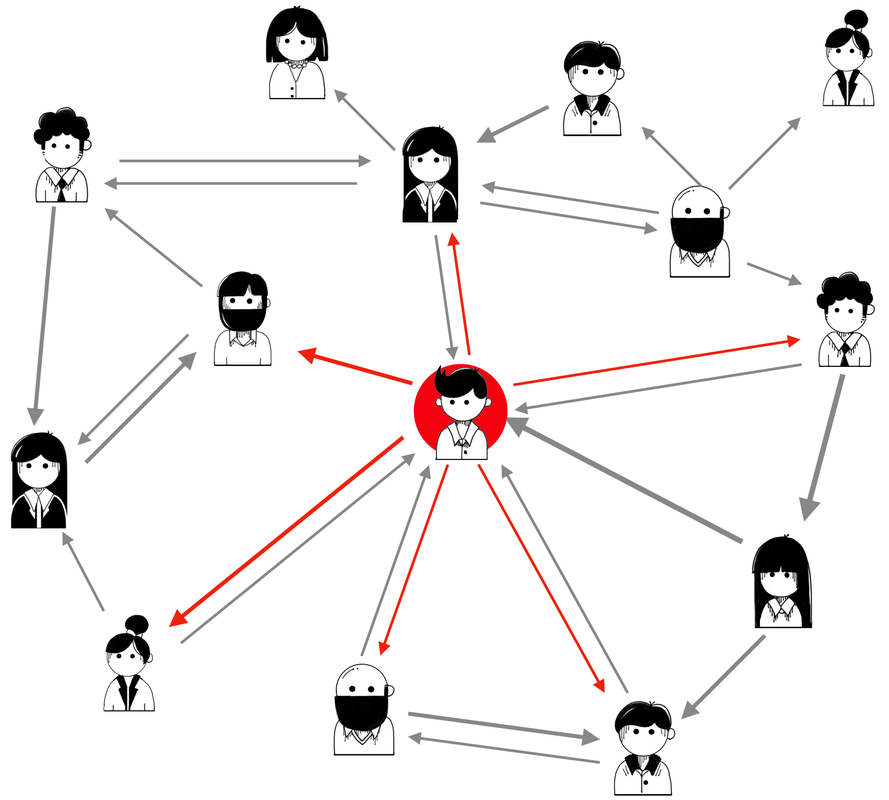
- Every agent i is endowed with a **payoff function** $V_i(a_i, \mathbf{a}_{-i})$ and is looking for

$$a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i(a_i, \mathbf{a}_{-i})$$



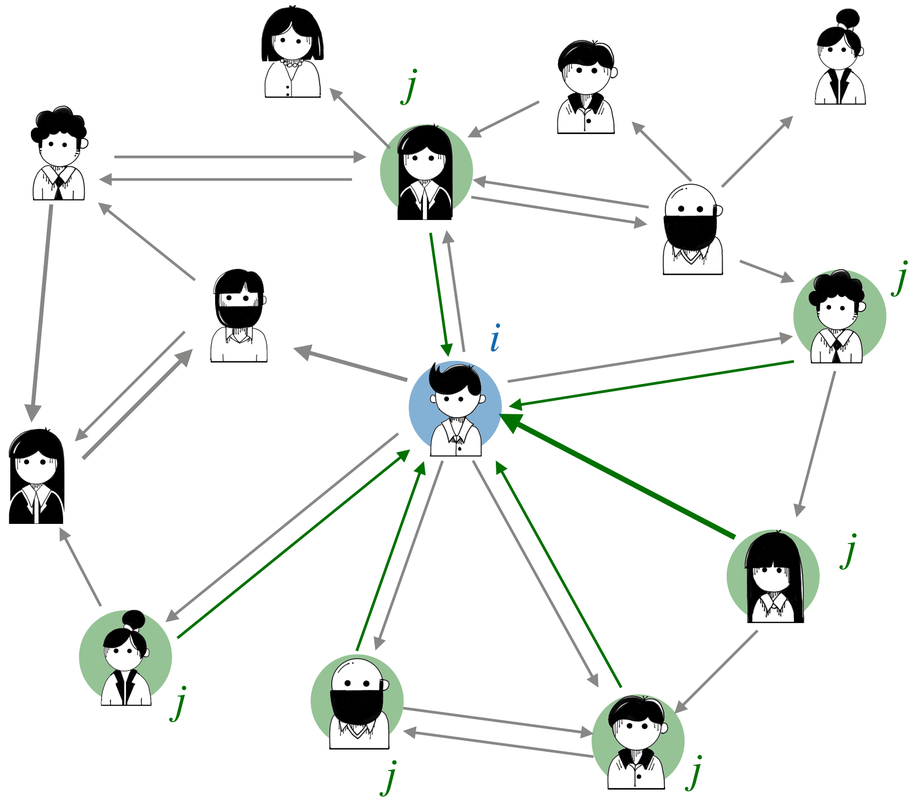
MODEL

$$V_i(\mathbf{a}_{-i}) = t_i(\mathbf{a}_{-i})$$

PAYOFF FUNCTION

Social Influence on friends

$$t_i(\mathbf{a}_{-i}) = \sum_{j \neq i} a_{ji}$$



$$V_i(a_i, \mathbf{a}_{-i} \mid \delta_i) = t_i(a_i, \mathbf{a}_{-i} \mid \delta_i)$$

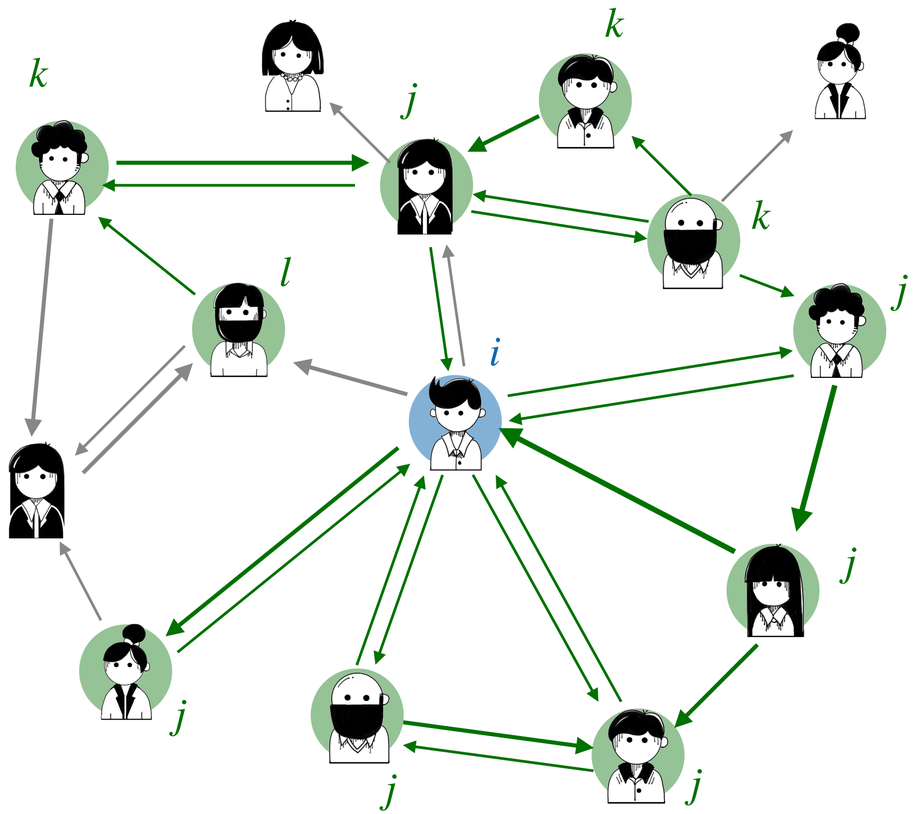
$$\delta_i \in [0,1]$$

PAYOFF FUNCTION

Social Influence on friends

$$t_i(a_i, \mathbf{a}_{-i} \mid \delta_i) = \sum_{j \neq i} a_{ji} + \underbrace{\delta_i \sum_{k \neq j} \sum_{j \neq i} a_{kj} a_{ji}}_{\text{paths of length 2}} + \underbrace{\delta_i^2 \sum_{l \neq k} \sum_{k \neq j} \sum_{j \neq i} a_{lk} a_{kj} a_{ji}}_{\text{paths of length 3}}$$

[Jackson, M. O. & Wolinsky, A. A strategic model of social and economic networks. J. Econom. Theory 71, 44–74 (1996).]



$$V_i(a_i, \mathbf{a}_{-i} \mid \delta_i) = t_i(a_i, \mathbf{a}_{-i} \mid \delta_i) + u_i(a_i, \mathbf{a}_{-i})$$

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Clustering Coefficient

$$u_i(a_i, \mathbf{a}_{-i}) = \sum_{j \neq i} a_{ij} \left(\sum_{k \neq i,j} a_{ik} a_{kj} \right)$$

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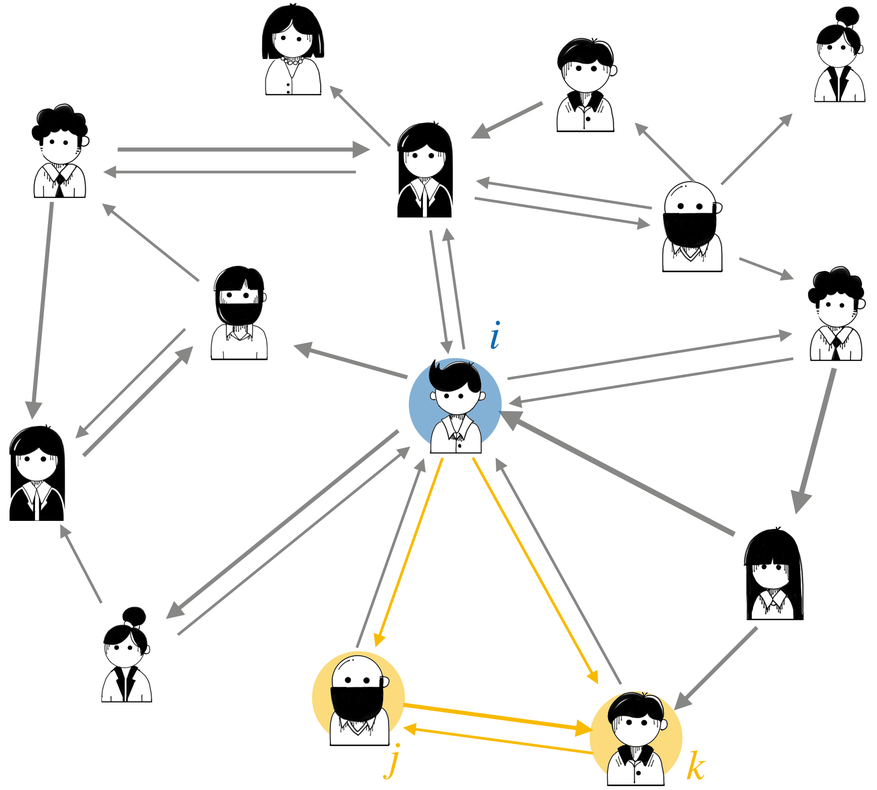

$$V_i(a_i, \mathbf{a}_{-i} \mid \delta_i) = t_i(a_i, \mathbf{a}_{-i} \mid \delta_i) + u_i(a_i, \mathbf{a}_{-i}) - c_i(a_i)$$

$$\delta_i \in [0,1]$$

PAYOFF FUNCTION

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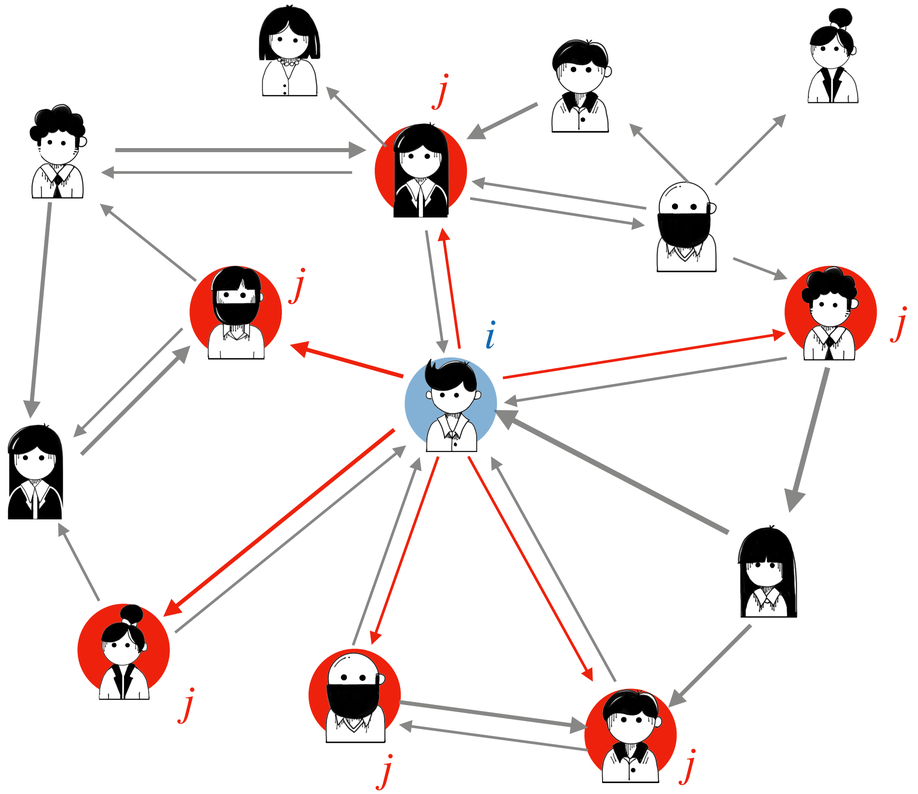
Cost $c_i(a_i) = \sum_{j \neq i} a_{ij}$



PAYOFF FUNCTION

$$V_i(a_i, \mathbf{a}_{-i} \mid \theta_i) = \alpha_i\, t_i(a_i, \mathbf{a}_{-i} \mid \delta_i) + \beta_i\, u_i(a_i, \mathbf{a}_{-i}) - \gamma_i\, c_i(a_i)$$

$$\theta_i = \{\alpha_i, \beta_i, \gamma_i, \delta_i\}$$

$$\alpha_i \geq 0, \beta_i \in \mathbb{R}, \gamma_i > 0, \delta_i \in [0,1]$$

SOCIAL INFLUENCE ON FRIENDS

$$t_i(a_i, \mathbf{a}_{-i} \mid \delta_i) = \sum_{j \neq i} a_{ji} + \underbrace{\delta_i \sum_{k \neq j} \sum_{j \neq i} a_{kj} a_{ji}}_{\text{paths of length 2}} + \underbrace{\delta_i^2 \sum_{l \neq k} \sum_{k \neq j} \sum_{j \neq i} a_{lk} a_{kj} a_{ji}}_{\text{paths of length 3}}$$

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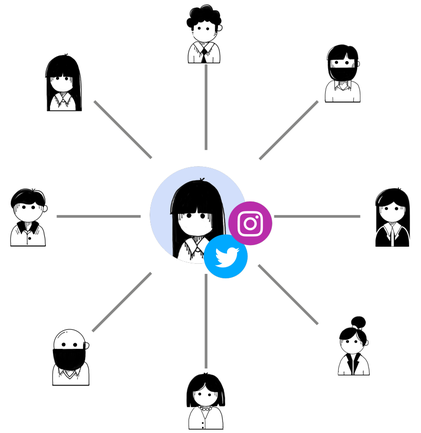
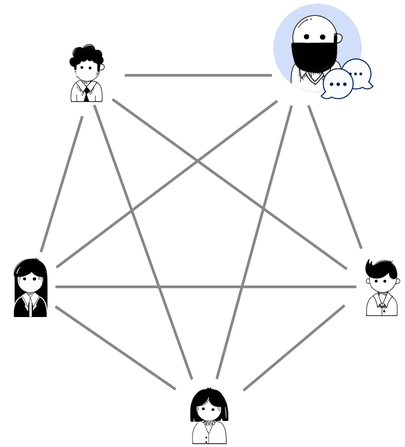
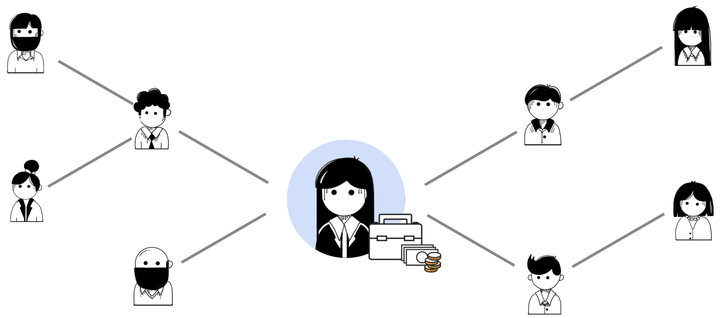


SOCIAL INFLUENCE

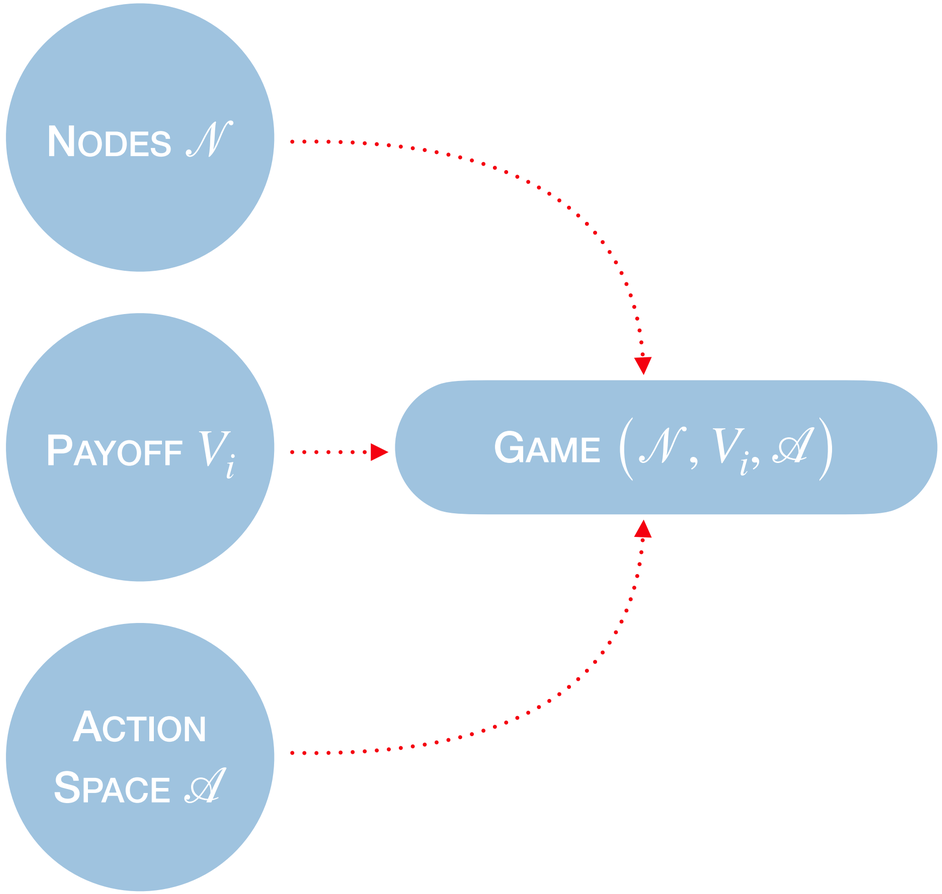


SOCIAL SUPPORT



BROKERAGE

GAME-THEORETICAL SETUP



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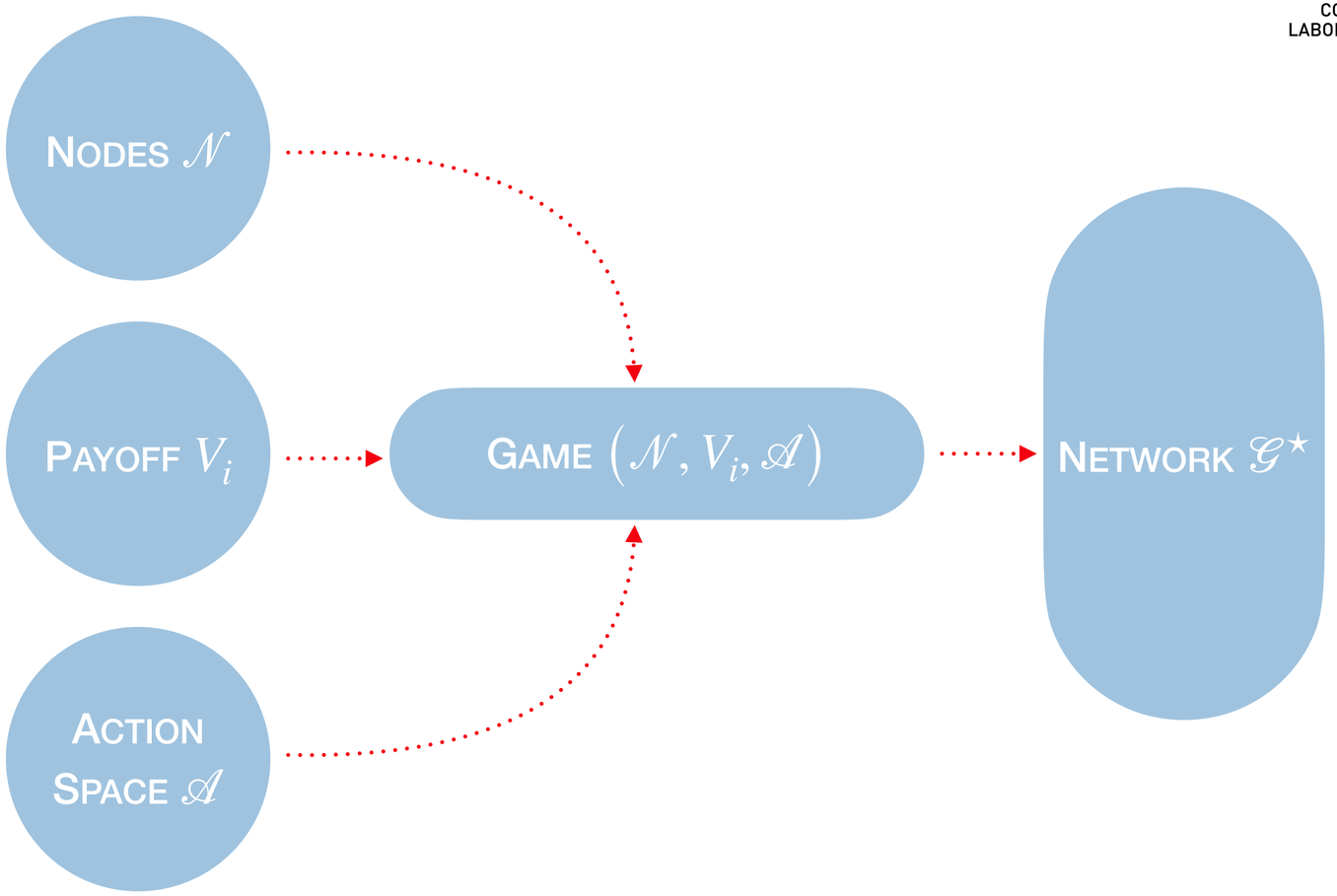


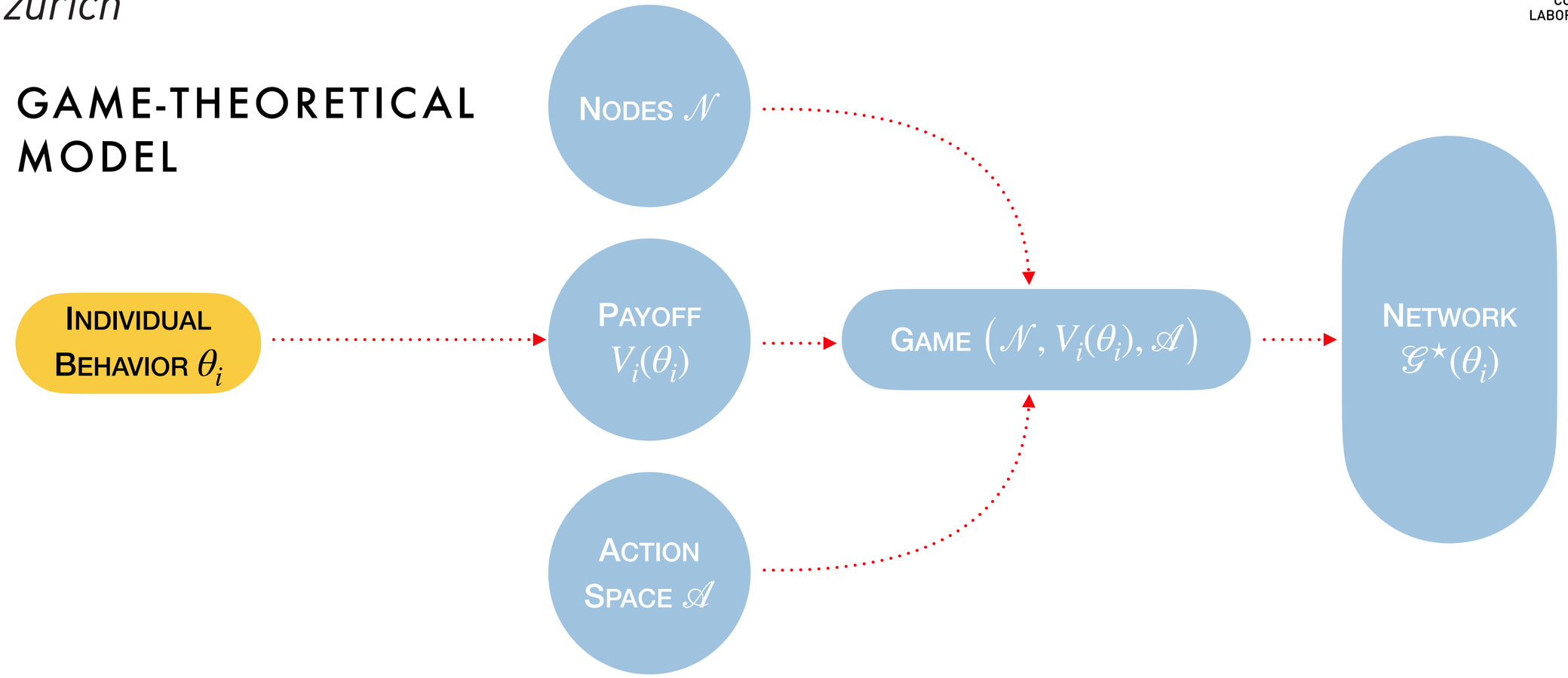
DEFINITION.

The network $\mathcal{G}^\star$ is a **Nash Equilibrium** if for all agents i

$$V_i\left(a_i, \mathbf{a}_{-i}^\star | \theta_i\right) \leq V_i\left(a_i^\star, \mathbf{a}_{-i}^\star | \theta_i\right), \forall a_i \in \mathcal{A}$$

$$\Longrightarrow \quad \forall i,\, a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star | \theta_i\right)$$

GAME-THEORETICAL MODEL



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$$\Longrightarrow \quad \forall i,\ a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}^\star_{-i} | \theta_i\right)$$

QUESTION: Given θ_i, which $\mathcal{G}^\star$ is in equilibrium ?

INDIVIDUAL BEHAVIOR θ_i

STRATEGIC NETWORK FORMATION MODEL

DETERMINE

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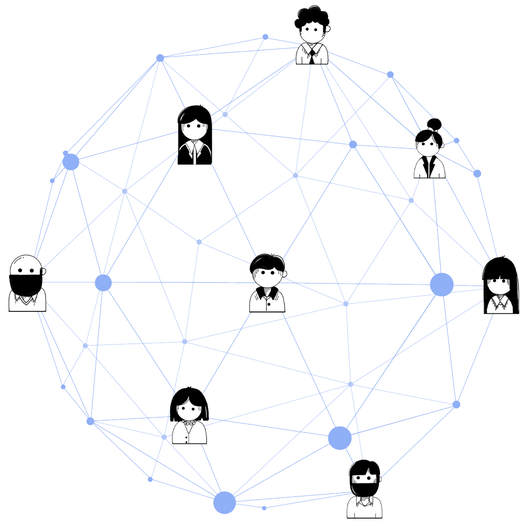

SOCIAL NETWORK STRUCTURE $\mathcal{G}^\star\left(\theta_i\right)$

MODEL

LITERATURE REVIEW

ASSUMPTION: Single incentive, homogeneous behavior θ_i

INDIVIDUAL BEHAVIOR θ_i

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BROKERAGE

DETERMINE

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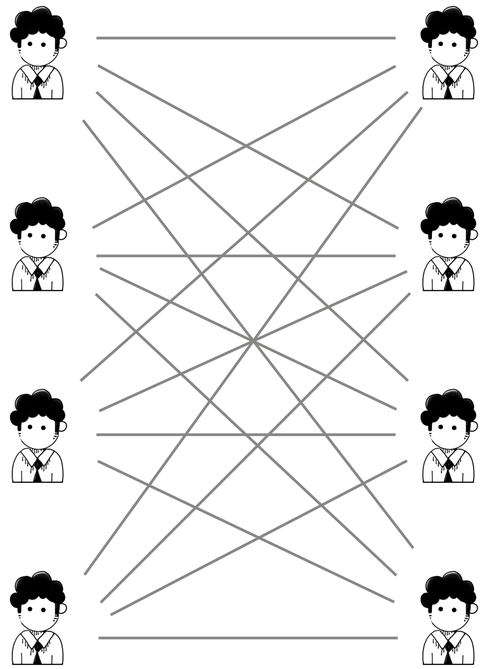

[Buechel, B. & Buskens, V. The dynamics of closeness and betweenness. J.Math. Sociol. 37(), 159–191 (2013)]

ASSUMPTION: Single incentive, homogeneous behavior θ_i

INDIVIDUAL BEHAVIOR θ_i

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DETERMINE

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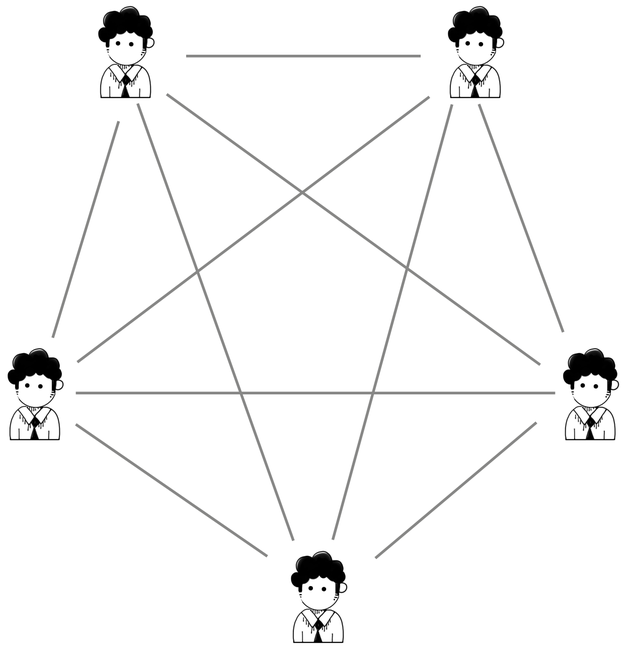
SOCIAL SUPPORT



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LITERATURE REVIEW

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DETERMINE

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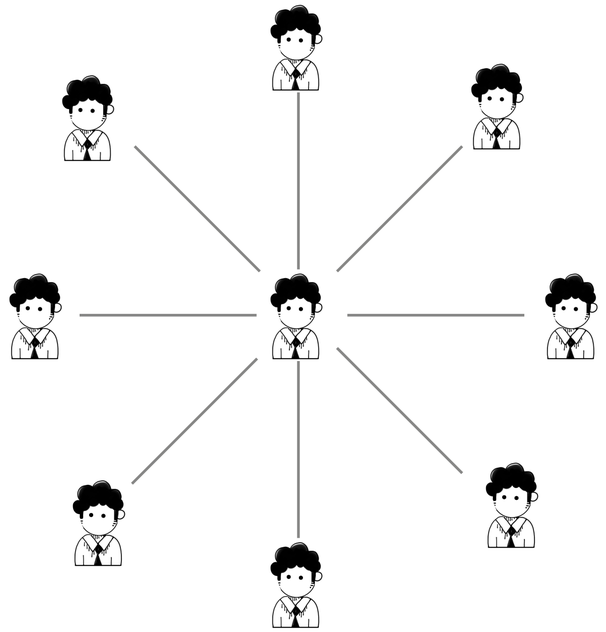


SOCIAL INFLUENCE

[Buechel, B. In Networks, Topology and Dynamics. Springer Lecture Notes in Economic and Mathematical Systems Vol. 613, 95–109 (Springer, 2008)]

LITERATURE REVIEW

RESULTS

HOMOGENEOUS RATIONAL AGENTS

$$V_i(a_i, \mathbf{a}_{-i} \mid \theta) = \frac{\alpha}{\gamma} t_i(a_i, \mathbf{a}_{-i}, \delta) + \frac{\beta}{\gamma} u_i(a_i, \mathbf{a}_{-i}) - c_i(a_i)$$

$$\theta = \{\alpha, \beta, \gamma, \delta\}$$

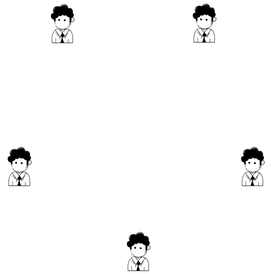
$$\alpha \geq 0, \beta \in \mathbb{R}, \gamma > 0, \delta \in [0,1]$$

ASSUMPTIONS.

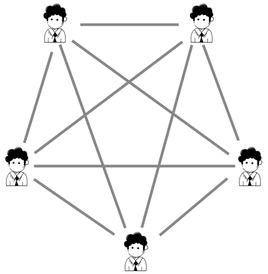
(i) **Homogeneity**: $\theta_i = \theta$, for all agents i.

(ii) Fully **rational** agents.

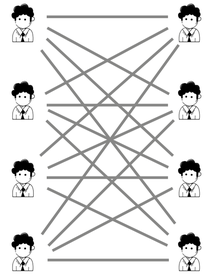
We consider 4 different stylised network motifs, and we derive **Necessary** and **Sufficient** conditions for Nash equilibrium stability through the **Variational Inequality** approach.



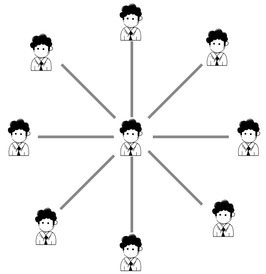
Empty



Complete



Complete Balanced Bipartite



Star

RESULTS

ETH zürich

HOMOGENEOUS RATIONAL AGENTS

$$V_i(a_i, \mathbf{a}_{-i} \mid \theta) = \frac{\alpha}{\gamma} t_i(a_i, \mathbf{a}_{-i}, \delta) + \frac{\beta}{\gamma} u_i(a_i, \mathbf{a}_{-i}) - c_i(a_i)$$

$$\theta = \{\alpha, \beta, \gamma, \delta\}$$

$$\alpha \geq 0, \beta \in \mathbb{R}, \gamma > 0, \delta \in [0,1]$$

ASSUMPTIONS.

(i) **Homogeneity**: $\theta_i = \theta$, for all agents i.

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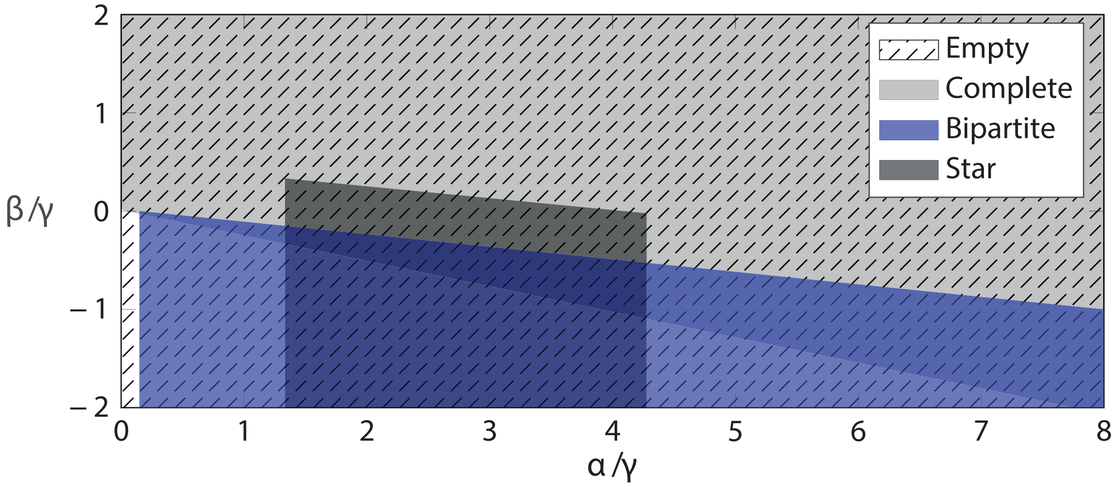
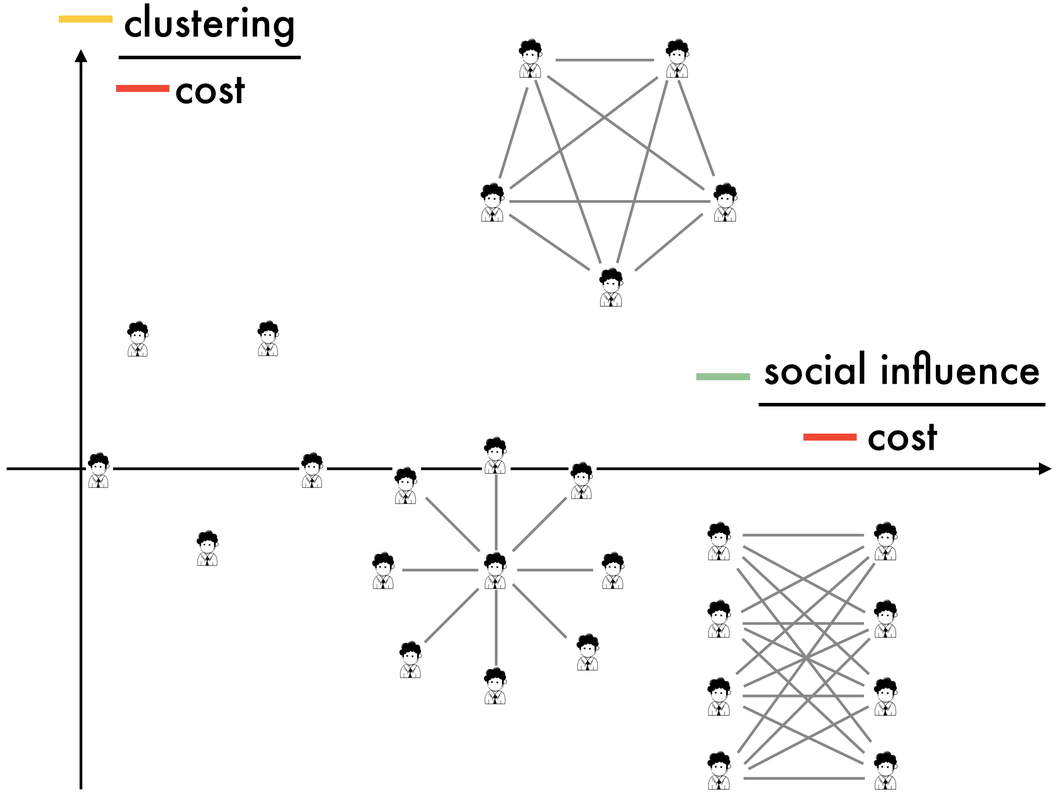




RESULTS

MODEL II

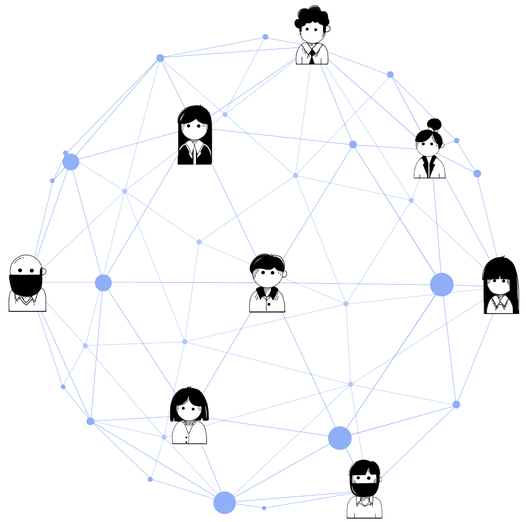
QUESTION: Given θ_i, which $\mathcal{G}^\star$ is in equilibrium ?

INDIVIDUAL BEHAVIOR θ_i

STRATEGIC NETWORK FORMATION MODEL

DETERMINE

$$\forall i,\ a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star \mid \theta_i\right)$$



SOCIAL NETWORK STRUCTURE $\mathcal{G}^\star\left(\theta_i\right)$

MODEL

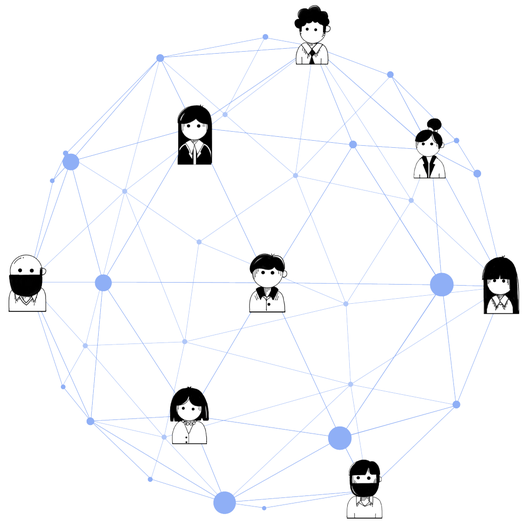
Question: Given $\mathcal{G}^\star$, for which θ_i is $\mathcal{G}^\star$ in equilibrium ?

INDIVIDUAL BEHAVIOR θ_i

STRATEGIC NETWORK FORMATION MODEL

DETERMINE

$$\forall i,\ a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star \mid \theta_i\right)$$



SOCIAL NETWORK STRUCTURE $\mathcal{G}^\star\left(\theta_i\right)$

STOCHASTIC ACTOR-ORIENTED MODEL

MODEL

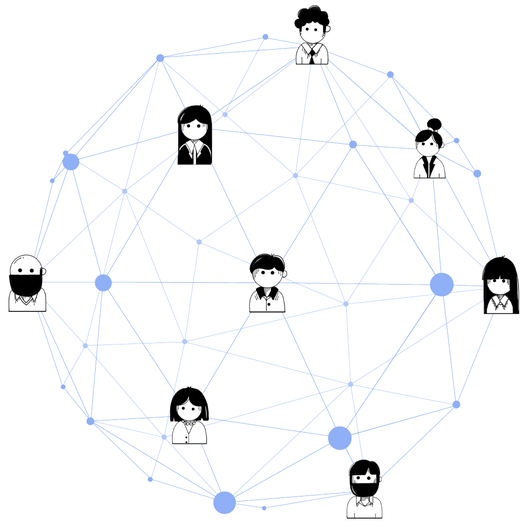
Question: Given $\mathcal{G}^\star$, for which θ_i is $\mathcal{G}^\star$ in equilibrium ?

INDIVIDUAL BEHAVIOR θ_i

STRATEGIC NETWORK FORMATION MODEL

DETERMINE

$$\forall i,\ a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star \,|\, \theta_i\right)$$



SOCIAL NETWORK STRUCTURE $\mathcal{G}^\star\left(\theta_i\right)$

GAME THEORETICAL INFERENCE

$$\forall i,\ \textbf{find}\ \ \theta_i\ \ \textbf{s.t.}\ \ V_i\left(a_i, \theta_i \,|\, \mathbf{a}_{-i}^\star\right) \leq V_i\left(\theta_i \,|\, a_i^\star, \mathbf{a}_{-i}^\star\right),\ \forall a_i \in \mathcal{A}$$

MODEL

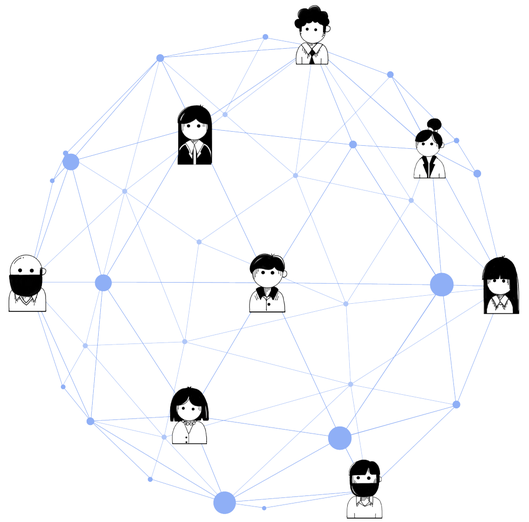
Question: Given $\mathcal{G}^\star$, for which θ_i is $\mathcal{G}^\star$ in equilibrium ?

INDIVIDUAL BEHAVIOR θ_i

STRATEGIC NETWORK FORMATION MODEL

DETERMINE

$$\forall i,\ a_i^\star \in \arg\max_{a_i \in \mathcal{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star \mid \theta_i\right)$$



SOCIAL NETWORK STRUCTURE $\mathcal{G}^\star\left(\theta_i\right)$

GAME-THEORETICAL INFERENCE

θ_i providing the **most rational** explanation

INVERSE OPTIMIZATION PROBLEM

Error function.

Deviation from Nash equilibrium:

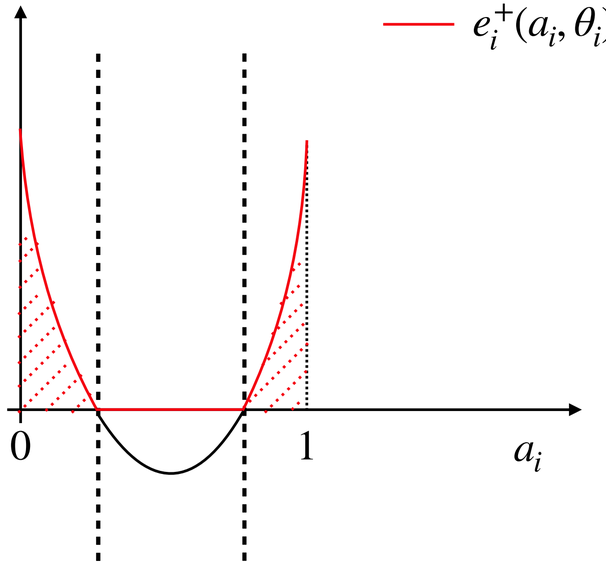
$$e_i(a_i, \theta_i) := V_i\left(a_i, \theta_i \,|\, \mathbf{a}^\star_{-i}\right) - V_i\left(\theta_i \,|\, a^\star_i, \mathbf{a}^\star_{-i}\right)$$

Positive error corresponds to a violation of the Nash equilibrium condition:

$$e_i^+(a_i, \theta_i) := \max\left\{0,\, e_i(a_i, \theta_i)\right\} > 0$$

Distance function.

$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i\right)^{1/2} = \|e_i^+(a_i, \theta_i)\|_{L_2(\mathscr{A})}$$



$e_i(a_i, \theta_i) < 0$

No violations: can be neglected

MODEL

INVERSE OPTIMIZATION PROBLEM

PROBLEM [MINIMUM NE-DISTANCE PROBLEM].

Given a network $\mathcal{G}^\star$ of N agents, for all agents i find the vectors of preferences $\theta_i^\star$ such that

$$\theta_i^\star \in \arg\min_{\theta_i \in \Theta} \; d_i(\theta_i)$$

INVERSE OPTIMIZATION PROBLEM

Problem [Minimum NE-Distance Problem].

Given a network $\mathcal{G}^\star$ of N agents, for all agents i find the vectors of preferences $\theta_i^\star$ such that

$$\theta_i^\star \in \arg\min_{\theta_i \in \Theta} \; d_i(\theta_i)$$

Theorem [Smoothness & convexity of distance function].

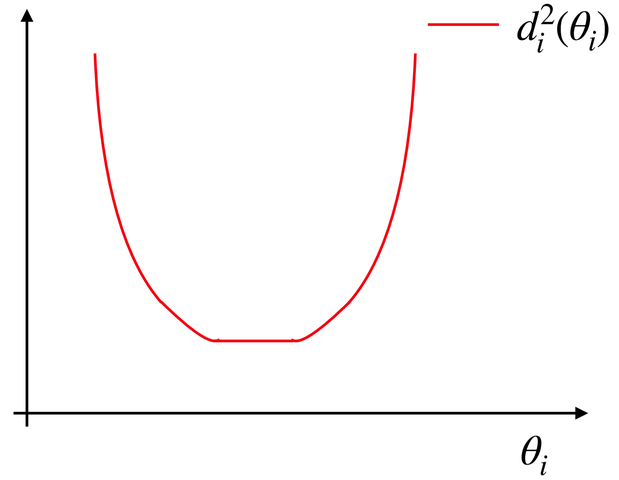
Let

$$d_i^2(\theta_i) := \int_{\mathscr{A}} \Big(\max \big\{ 0,\, e_i(a_i, \theta_i) \big\} \Big)^2 dx = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}^2$$

Then $d_i^2(\theta_i)$ is continuously differentiable, and its gradient reads as

$$\nabla_\theta d_i(\theta) = \int_{\mathscr{A}} 2\, \nabla_{\theta_i} \big(e_i(a_i, \theta_i) \big) \max \big\{ 0,\, e_i(a_i, \theta_i) \big\}\, da_i \,.$$

Moreover, $d_i^2(\theta_i)$ is convex.

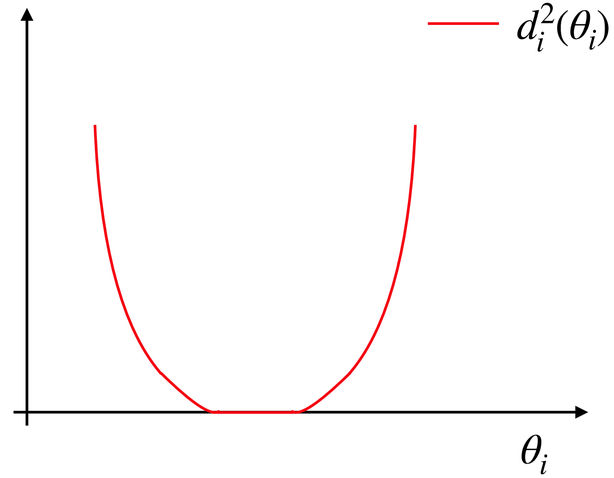


$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.



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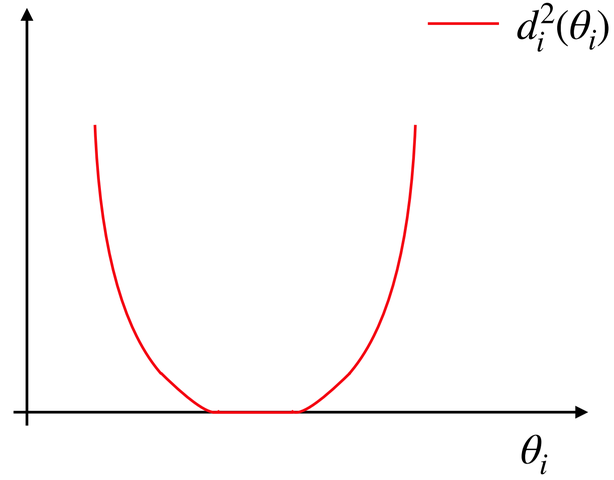

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Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

First, note that, since $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$, then $\Theta_{i,0} := \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}, e_i(a_i, \theta_i) \leq 0 \right\}$.

$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

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Observe that $e_i(a_i, \theta_i) = x_i(a_i)\theta_i - y_i(a_i)$ is a combination of linear and mixed quadratic terms in the components of a_i, but there are not no pure quadratic terms of the form a_{ij}^2.

$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

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$$Hess_{a_i}\left(e_i(a_i, \theta_i)\right) = \begin{bmatrix} 0 & \cdot & \cdot & \cdot & \cdot \\ \cdot & 0 & \cdot & \cdot & \cdot \\ & & \ddots & & \\ \cdot & \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & \cdot & 0 \end{bmatrix}$$

Then $Trace\left(Hess_{a_i} e_i(a_i, \theta_i)\right) = 0$, i.e., $\sum_{j=1}^{N-1} \lambda_j = 0$.

Thus, any critical point in the compact set $\mathscr{A}$ is a saddle point, and cannot be a strict maximum.

$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

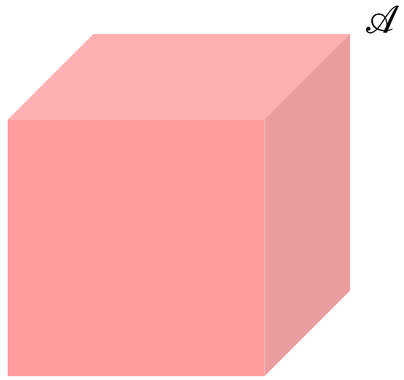
INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

Then the maximum of $\max_{a_i \in \mathscr{A}} e_i(a_i, \theta_i)$ must be attained on the boundary of $\mathscr{A}$, which is composed by $2(N-1)$ sides, each side being the compact set $[0,1]^{N-2}$.



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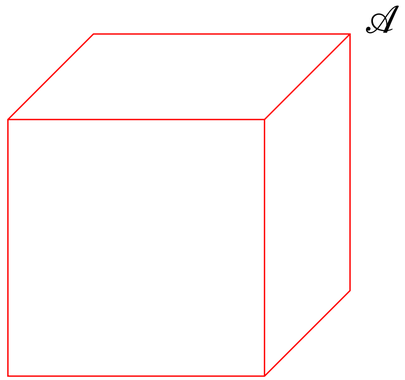
INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

On the side, the error function has the same property.
By induction...



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INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

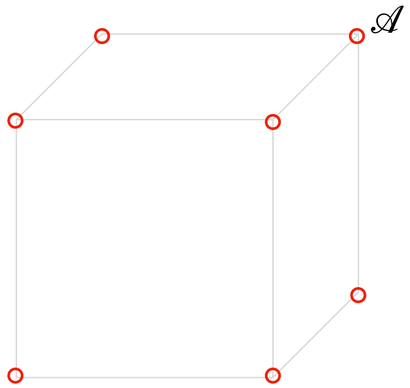
Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

Finally...

$\Theta_{i,0} := \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}_{\{0,1\}}, e_i(a_i, \theta_i) \leq 0 \right\}$,

where $\mathscr{A}_{\{0,1\}} = \{0, 1\}^{N-1}$.



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INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

$\Theta_{i,0} = \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}_{\{0,1\}}, x_i(a_i)\theta_i - y_i(a_i) \leq 0 \right\}$, where $\mathscr{A}_{\{0,1\}} = \{0, 1\}^{N-1}$.

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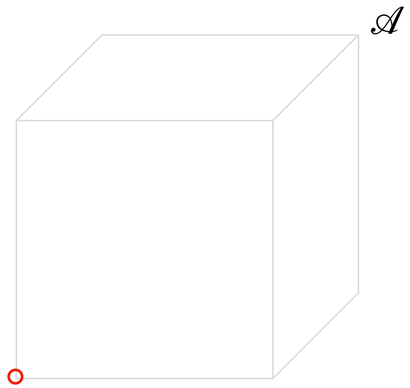
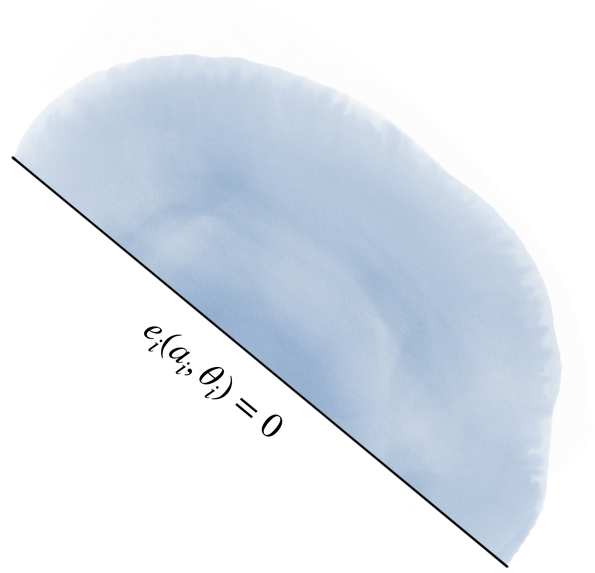




$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

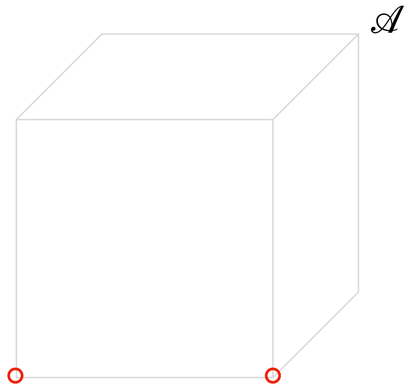
INVERSE OPTIMIZATION PROBLEM - SOLUTION

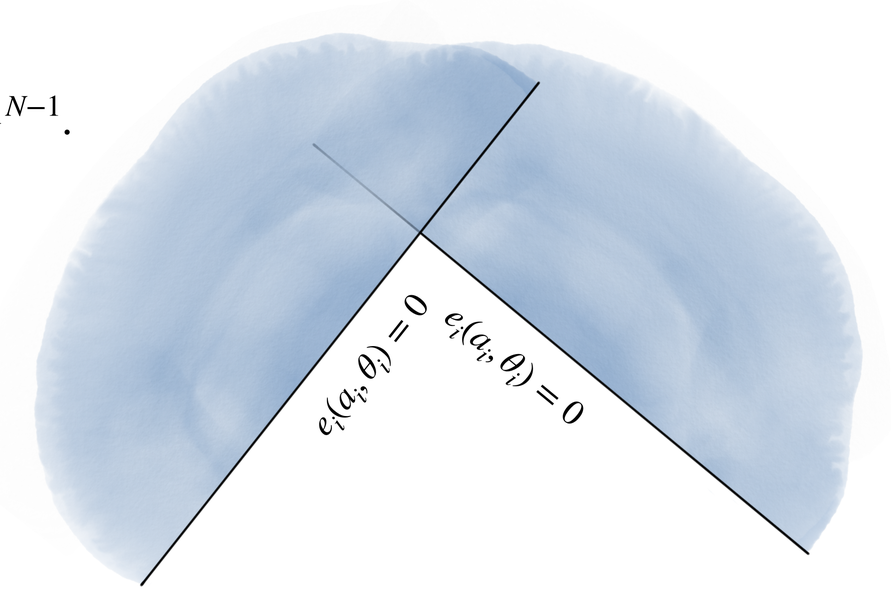
THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

$\Theta_{i,0} = \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}_{\{0,1\}}, x_i(a_i)\theta_i - y_i(a_i) \leq 0 \right\}$, where $\mathscr{A}_{\{0,1\}} = \{0, 1\}^{N-1}$.





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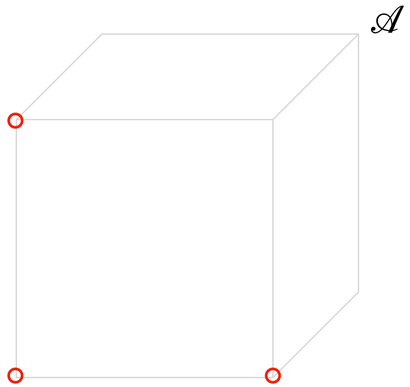
INVERSE OPTIMIZATION PROBLEM - SOLUTION

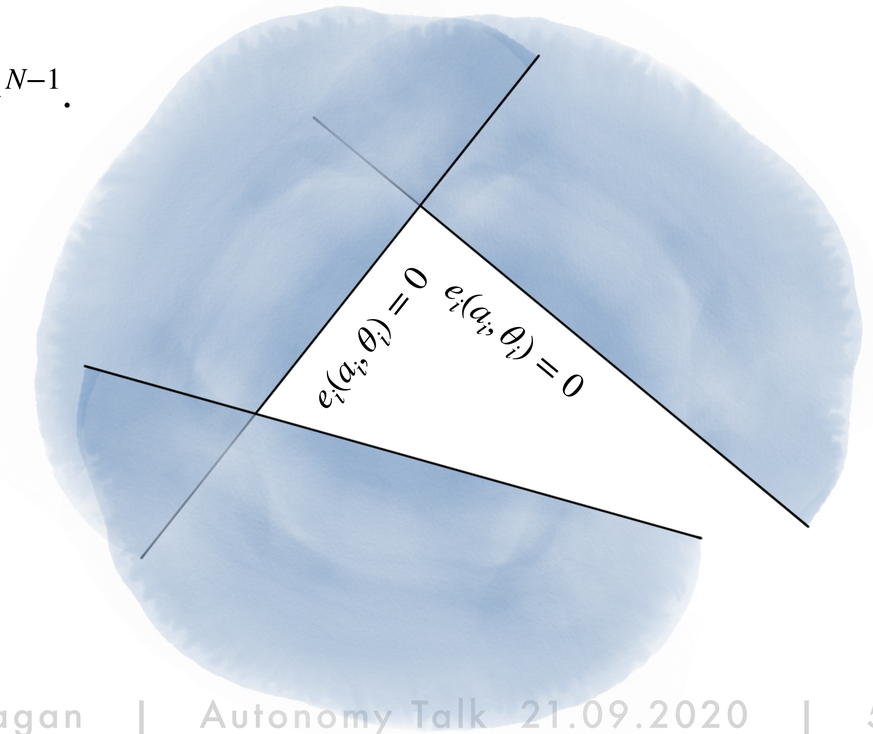
THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

$\Theta_{i,0} = \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}_{\{0,1\}}, x_i(a_i)\theta_i - y_i(a_i) \leq 0 \right\}$, where $\mathscr{A}_{\{0,1\}} = \{0, 1\}^{N-1}$.





$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

THEOREM.

Assume $\min_{\theta_i \in \Theta} d_i^2(\theta_i) = 0$. Then, the set of solutions $\Theta_{i,0}$ is a polyhedron.

PROOF (SKETCH).

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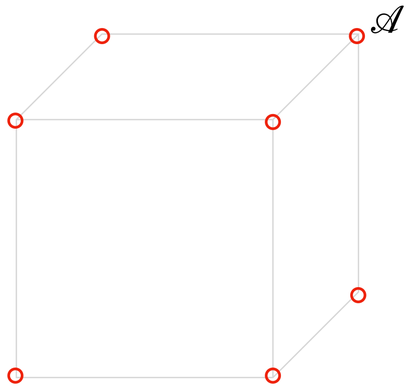

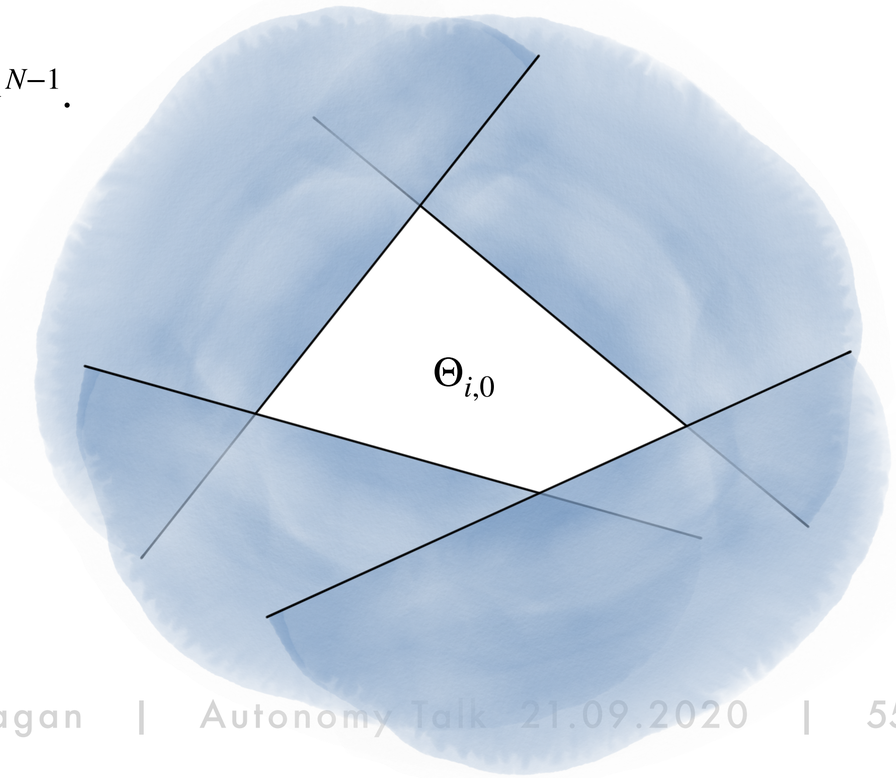

$$d_i(\theta_i) := \left(\int_{\mathscr{A}} e_i^+(a_i, \theta_i)^2 da_i \right)^{1/2} = \| e_i^+(a_i, \theta_i) \|_{L_2(\mathscr{A})}$$

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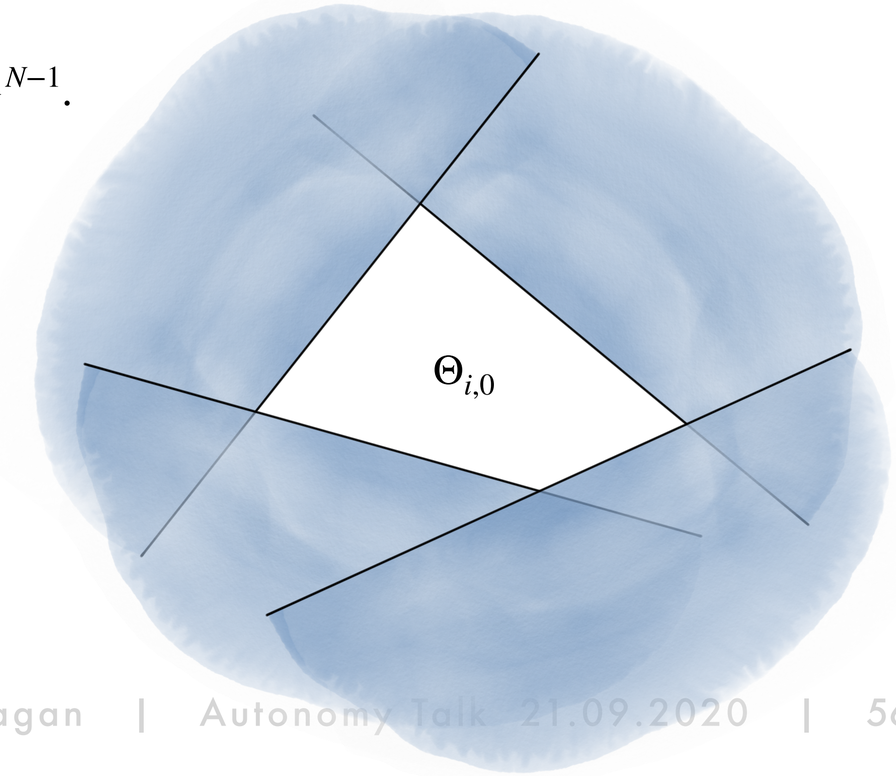
$\Theta_{i,0} = \left\{ \theta_i \in \Theta, \text{ s.t. } \forall a_i \in \mathscr{A}_{\{0,1\}}, x_i(a_i)\theta_i - y_i(a_i) \leq 0 \right\}$, where $\mathscr{A}_{\{0,1\}} = \{0, 1\}^{N-1}$.

Thus, if a set of exact solutions to the NE conditions exists, it can be found as the intersection of **finitely** many half-spaces.



INVERSE OPTIMIZATION PROBLEM - SOLUTION

If $\Theta_{i,0} = \varnothing$*?*

First-order optimality condition

$$0 = \nabla_{\theta_i}\left(d_i^2(\theta_i)\right) = 2\int_{\mathcal{A}} \nabla_{\theta_i}\left(e_i(a_i,\theta_i)\right) \max\left\{0,\, e_i(a_i,\theta_i)\right\} da_i\,.$$

max operator within $(N-1)$ - dimensional integral

INVERSE OPTIMIZATION PROBLEM - SOLUTION

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max operator within $(N-1)$ - dimensional integral

Removing the max operator, integral over $\mathscr{A}_+(\theta_i) = \left\{a_i \in \mathscr{A} \,|\, e_i(a_i,\theta_i) > 0\right\}$,

$$0 = \nabla_{\theta_i}\left(d_i^2(\theta_i)\right) = 2\int_{\mathscr{A}_+(\theta_i)} \nabla_{\theta_i}\left(e_i(a_i,\theta_i)\right) e_i(a_i,\theta_i) da_i,$$

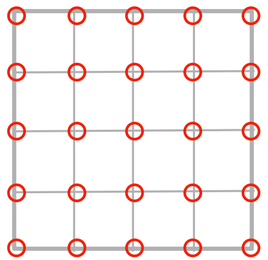
cannot be solved in closed form for θ_i (in the general case).

INVERSE OPTIMIZATION PROBLEM - SOLUTION

Search for an approximate solution. Consider a finite set of possible actions (samples)

$$\left\{ a_i^j \right\}_{j=1}^{n_i} \subset \mathscr{A}$$

Let $e_i^j(\theta_i) = e_i(a_i^j, \theta_i)$ and $e_i^{j,+}(\theta_i) = e_i^+(a_i^j, \theta_i)$ be the corresponding error and positive error at the samples.



INVERSE OPTIMIZATION PROBLEM - SOLUTION

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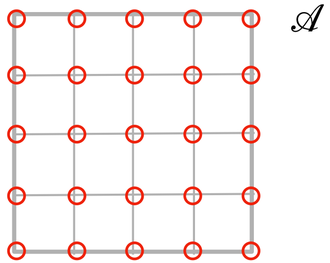

Let $e_i^j(\theta_i) = e_i(a_i^j, \theta_i)$ and $e_i^{j,+}(\theta_i) = e_i^+(a_i^j, \theta_i)$ be the corresponding error and positive error at the samples.

- Approximate the distance function as

$$\tilde{d}_i(\theta_i) := \left(\sum_{j=1}^{n_i} \left(e_i^{j,+}(\theta_i) \right)^2 \right)^{1/2} = \|\mathbf{e}_i^+\|_2$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

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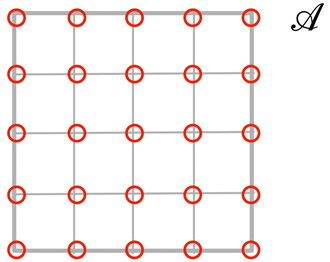

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$$\tilde{d}_i(\theta_i) := \left(\sum_{j=1}^{n_i} \left(e_i^{j,+}(\theta_i) \right)^2 \right)^{1/2} = \|\mathbf{e}_i^+\|_2$$

- *Problem [Discrete Minimum NE-Distance Problem].*
Given a network $\mathscr{G}^\star$ of N agents, for all agents i find the vectors of preferences $\hat{\theta}_i$ such that

$$\hat{\theta}_i \in \arg\min_{\theta_i \in \Theta} \ \tilde{d}_i(\theta_i)$$

QUESTION: How "good" is the approximate solution?

CONFIDENCE INTERVALS

PROBLEM [DISCRETE MINIMUM NE-DISTANCE PROBLEM].

Given a network $\mathcal{G}^\star$ of N agents, for all agents i find the vectors of preferences $\hat{\theta}_i$ such that

$$\hat{\theta}_i \in \arg\min_{\theta_i \in \Theta} \|\mathbf{e}_i^+(\theta_i)\|^2$$

where $e_i{}^+ = \max\left\{0, \mathbf{X_i}\theta - y\right\}$ is the vector of positive errors. Then

$$\hat{\theta}_i = \left(\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{X}_i(\hat{\theta}_i)\right)^{-1}\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{y}_i(\hat{\theta}_i)$$

Since the residuals are all non negative, we assume $\varepsilon_i \sim Exp\left(\lambda_i \mathbf{I}_n\right)$ and i.i.d..
Bias and confidence interval can still be derived.

PROBLEM [ORDINARY LEAST SQUARE (OLS) REGRESSION].

Given a set of n samples $\left\{x_i\right\}_{i=1}^n$ and n observations $\left\{y_i\right\}_{i=1}^n$, where each scalar y_i is the response to the row vector x_i of values of p predictors (regressors) x_{ij} for $j = 1, \ldots, p$, find $\hat{\theta} \in \mathbb{R}^p$ such that $\hat{\theta} = \arg\min_\theta \|e(\theta)\|_2^2$,

where $e = \mathbf{X}\theta - y$ is the vector of residuals. Then

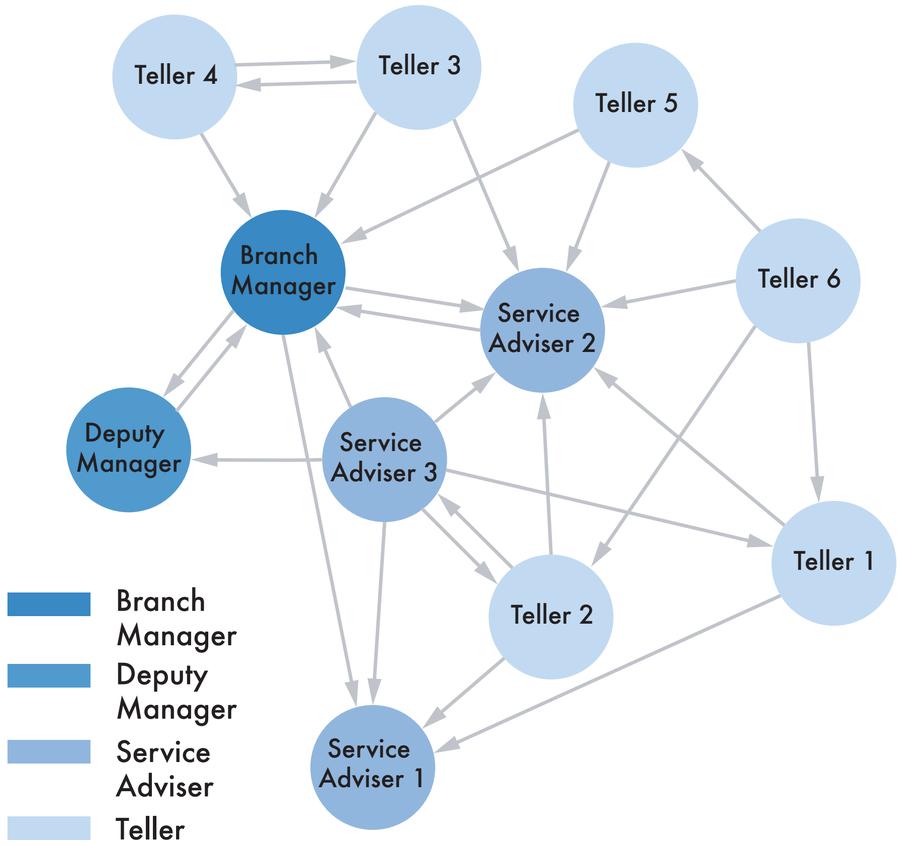
$$\hat{\theta} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}$$

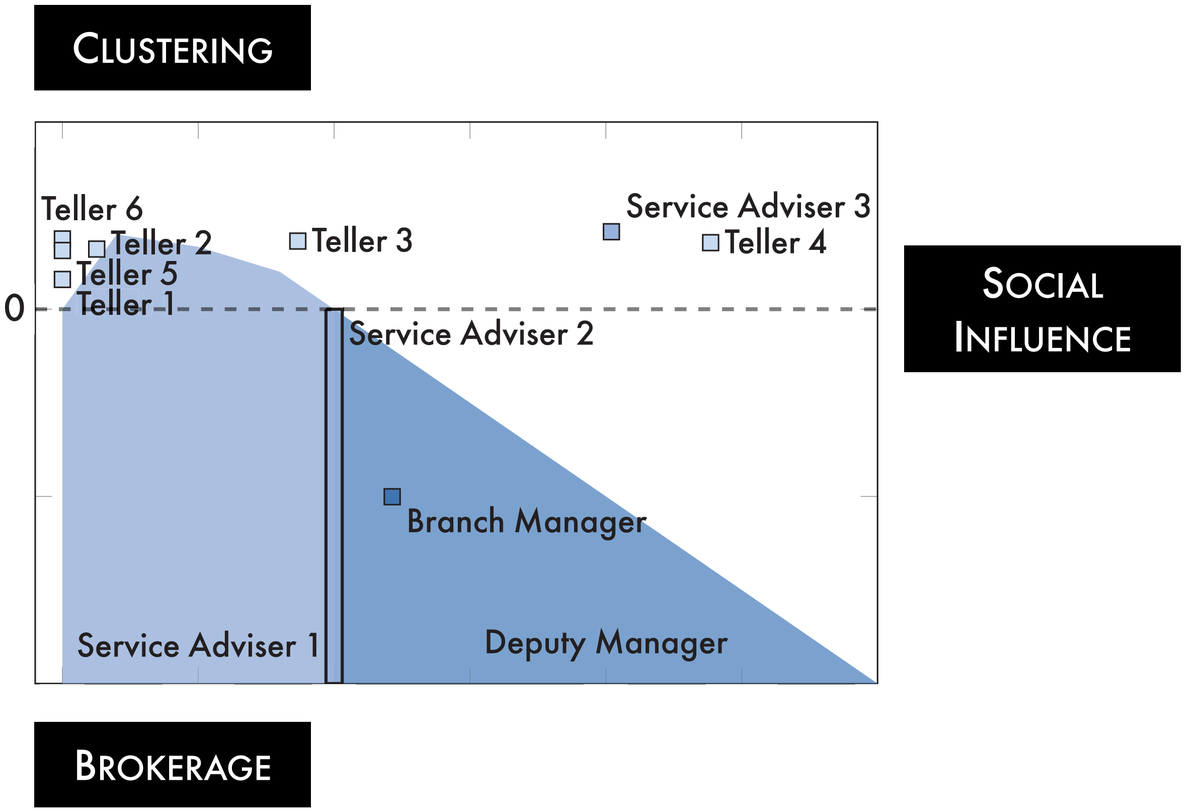
If $\varepsilon \sim \mathcal{N}\left(0, \sigma^2\mathbf{I}_n\right)$ and i.i.d, then the estimator $\hat{\theta}$ is unbiased, normally distributed, and its covariance matrix reads as

$$\sigma^2\mathbf{X}^T\mathbf{X}$$

RESULTS II

AUSTRALIAN BANK DATA-SET

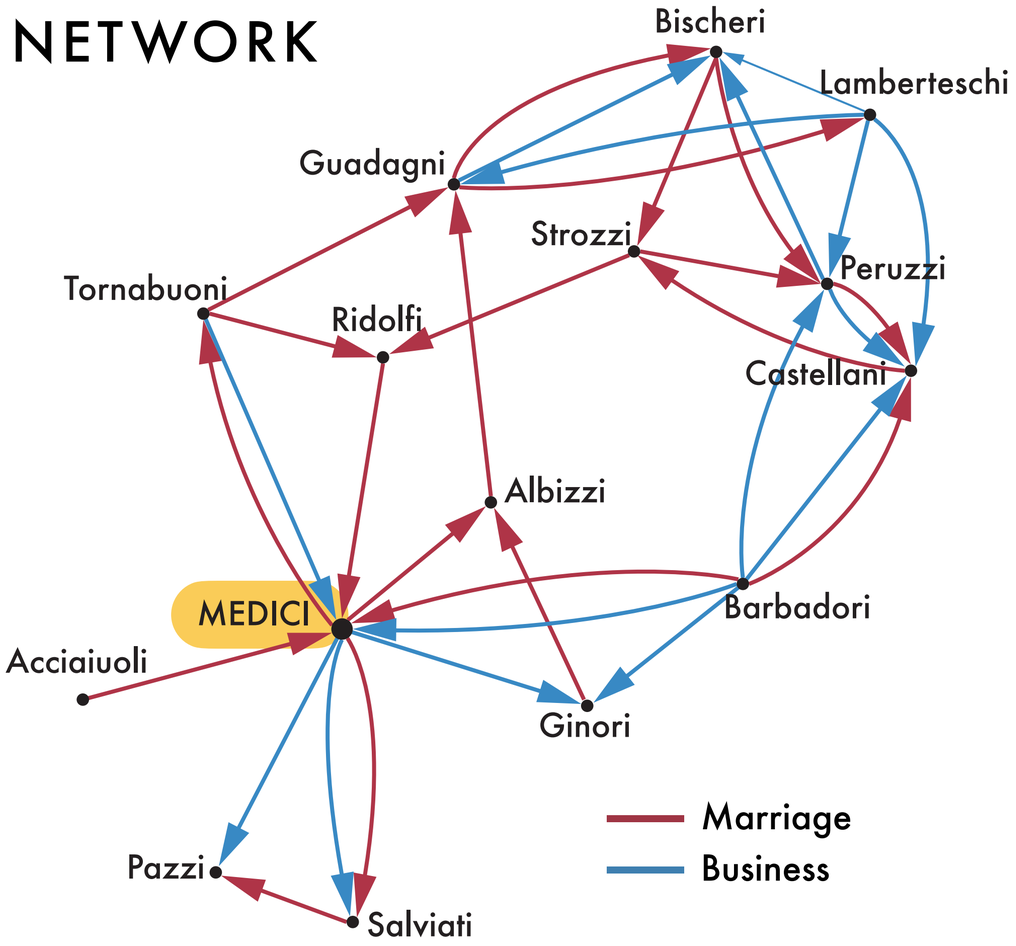


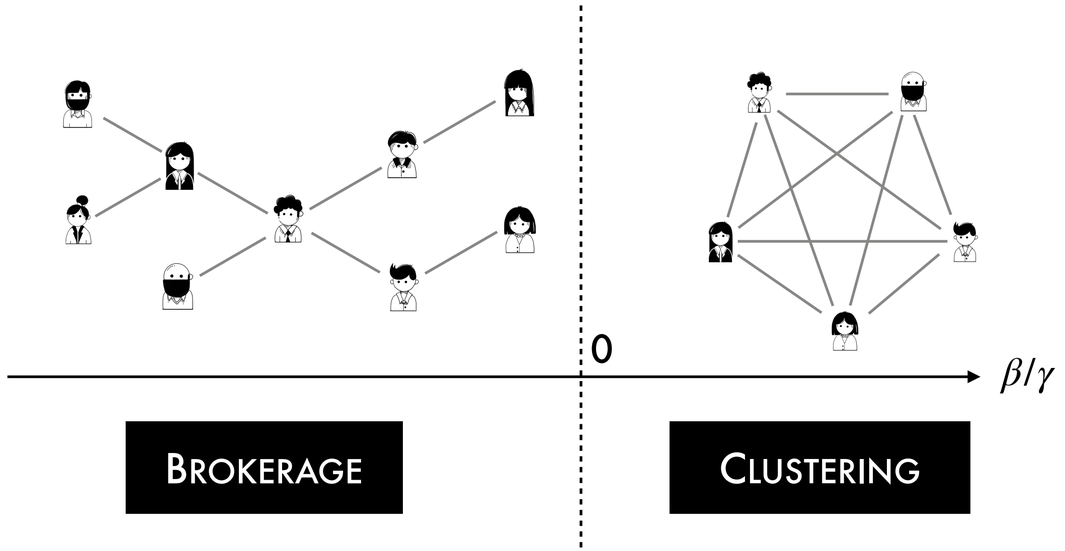


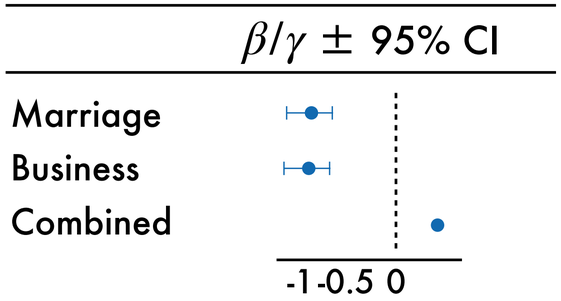
Pattison, P., Wasserman, S., Robins, G. & Kanfer, A. M. Statistical evaluation of algebraic constraints for social networks. J. Math. Psychology 44, (2000).

RESULTS

MEDICI NETWORK



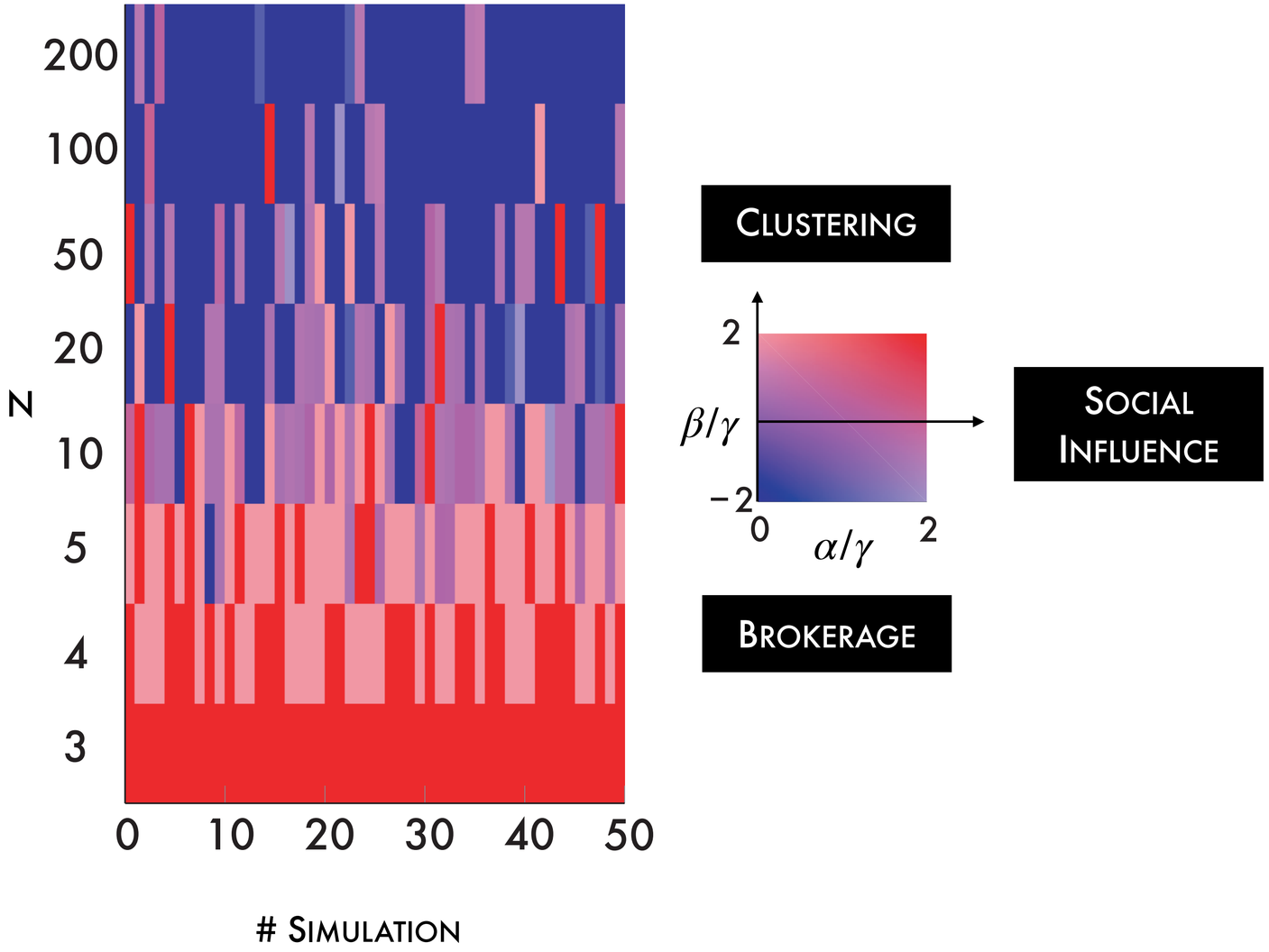




Padgett, J. F., & Ansell, C. K. (1993). Robust Action and the Rise of the Medici, 1400-1434. *American journal of sociology*, 98(6), 1259-1319.

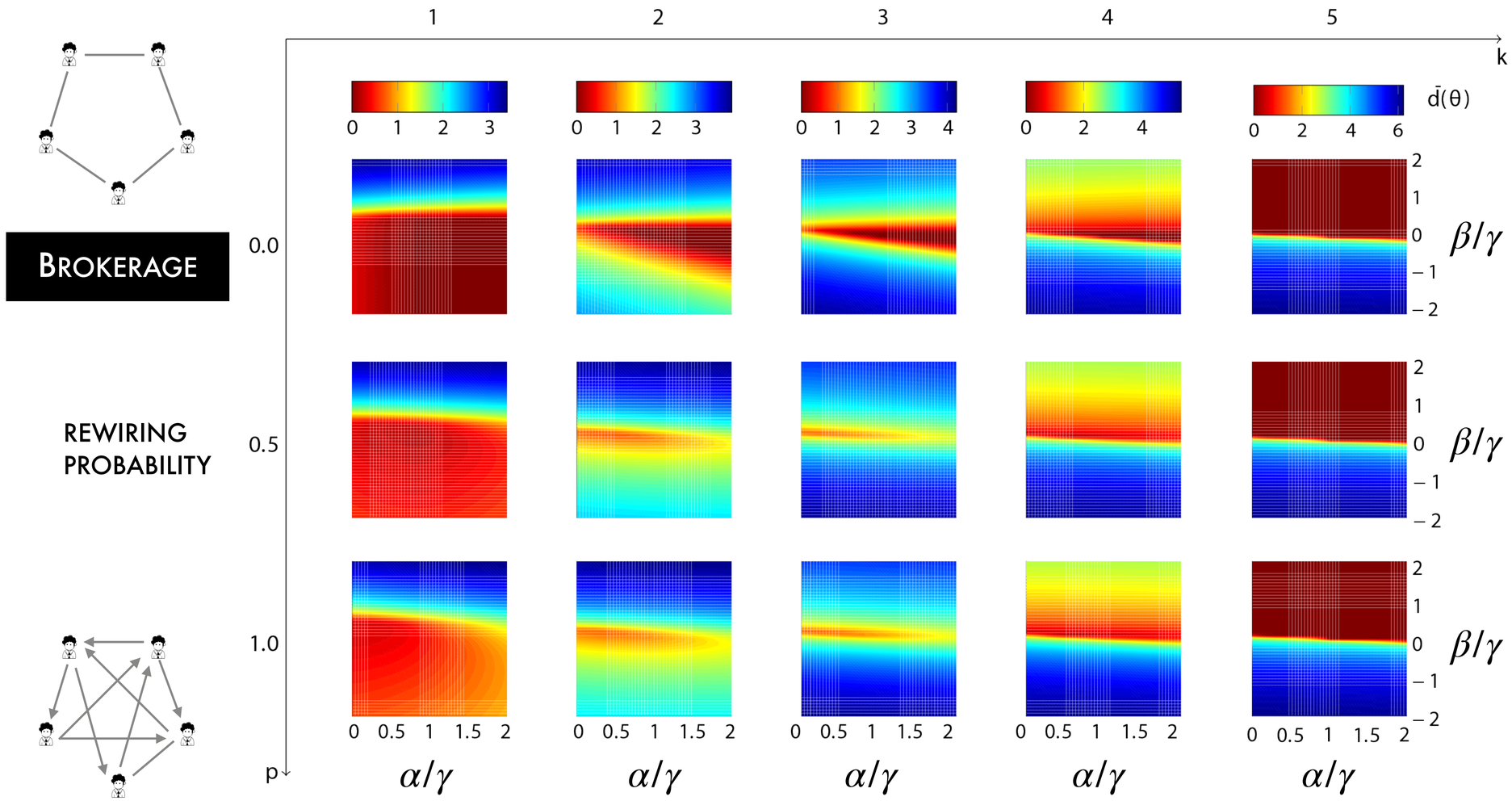
PREFERENTIAL ATTACHMENT

- Nodes are introduced sequentially.
  Each newborn **receives 2 incoming links** from existing nodes (randomly selected, proportionally to the outdegree), and **creates 2 outgoing ties** to existing nodes (randomly selected, proportionally to the indegree).



RESULTS

SMALL-WORLD NETWORK



CONCLUSIONS

SUMMARY & OPEN DIRECTIONS

- Starting from the strategic network formation literature, we proposed a new model:
  - sociologically well-founded,
  - mathematically tractable, and
  - statistically robust,

  capable to reverse-engineering human behavior from easily accessible data on the network structure.
- We provided evidence that our results are consistent with empirical, historical, and sociological observations.
- Our method offers socio-strategic interpretations of random network models.

- The model can be adapted to further specifications of the payoff function.
- Incorporating prior knowledge on the action space of the agents can reduce the computational burden.
- Actors' attributes have not yet been considered.

nature COMMUNICATIONS

Pagan, N. & Dörfler, F. *Game theoretical inference of human behavior in social networks.* NATURE COMMUNICATIONS, 2019.

NICOLÒ PAGAN
FLORIAN DÖRFLER

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ETH zürich

BACK UP SLIDES

HOMOGENEOUS RATIONAL AGENTS

$$V_i(a_i, \mathbf{a}_{-i} \,|\, \theta) = \frac{\alpha}{\gamma}\, t_i(a_i, \mathbf{a}_{-i}, \delta) + \frac{\beta}{\gamma}\, u_i(a_i, \mathbf{a}_{-i}) - c_i(a_i)$$

$$\theta = \{\alpha, \beta, \gamma, \delta\}$$

$$\alpha \geq 0, \beta \in \mathbb{R}, \gamma > 0, \delta \in [0,1]$$

Assumptions.

(i) **Homogeneity**: $\theta_i = \theta$, for all agents i.

(ii) Fully **rational** agents.

Definition.

The network $\mathcal{G}^\star$ is a **Nash Equilibrium** if for all agents i:

$$V_i\left(a_i, \mathbf{a}_{-\mathbf{i}}{}^\star \,|\, \theta\right) \leq V_i\left(a_i^\star, \mathbf{a}_{-\mathbf{i}}{}^\star \,|\, \theta\right), \quad \forall a_i \in \mathscr{A} \qquad \Longrightarrow \qquad \forall i,\, a_i^\star \in \arg\max_{a_i \in \mathscr{A}} V_i\left(a_i, \mathbf{a}_{-i}^\star \,|\, \theta\right).$$

Using the **Variational Inequality** approach, it is equivalent to

$$\left\langle \nabla V_i\left(a_i, \mathbf{a}_{-i}^\star \,|\, \theta\right)\Big|_{a_i^\star}^{T},\, a_i - a_i{}^\star \right\rangle \leq 0, \quad \forall a_i \in \mathscr{A}\,.$$

RESULTS

HOMOGENEOUS RATIONAL AGENTS

$$V_i(a_i, \mathbf{a}_{-i} \mid \theta) = \frac{\alpha}{\gamma}\, t_i(a_i, \mathbf{a}_{-i}, \delta) + \frac{\beta}{\gamma}\, u_i(a_i, \mathbf{a}_{-i}) - c_i(a_i)$$

$$\theta = \{\alpha, \beta, \gamma, \delta\}$$

$$\alpha \geq 0, \beta \in \mathbb{R}, \gamma > 0, \delta \in [0,1]$$

Assumptions.

(i) **Homogeneity**: $\theta_i = \theta$, for all agents i.

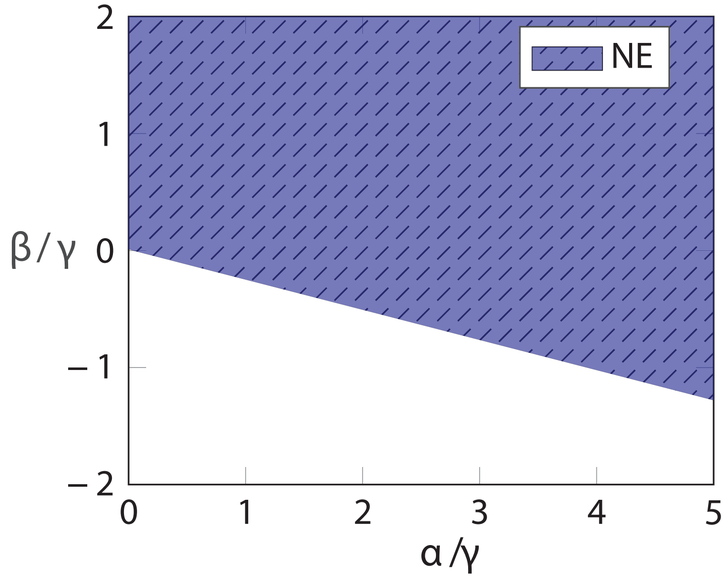
(ii) Fully **rational** agents.

Theorem.

Let $\mathcal{G}^{CN}$ be a complete network of N homogeneous, rational agents. Define:

$$\bar{\gamma}_{NE} := \begin{cases} \alpha\delta\left(1 + \delta(2N-3)\right) + \beta\,(N-2), & \text{if } \beta > 0 \\ \alpha\delta\left(1 + \delta(2N-3)\right) + 2\beta\,(N-2), & \text{if } \beta \leq 0, \end{cases}$$

then $\mathcal{G}^{CN}$ is a Nash equilibrium if and only if $\gamma \leq \bar{\gamma}_{NE}$.

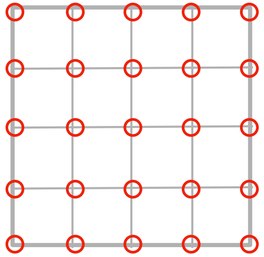


INVERSE OPTIMIZATION PROBLEM - SOLUTION

The first-order optimality condition $\nabla_{\theta_i}\left(\tilde{d}_i^{\,2}(\theta_i)\right) = 2\sum_{j=1}^{n_i} \nabla_{\theta_i} e_i^j(\theta_i) \max\left\{0,\, e_i^j(\theta_i)\right\} = 0$ can be written as:

$$\mathbf{X}_i^T \left(\max\left\{ \mathbf{0}_{n_i \times 1}, \mathbf{X}_i \theta_i - \mathbf{y}_i \right\} \right) = 0.$$

where



$\mathscr{A}$

samples

$$\underbrace{\begin{bmatrix} x_{i,1}(a_i^1) & x_{i,2}(a_i^1) & x_{i,3}(a_i^1) \\ x_{i,1}(a_i^2) & x_{i,2}(a_i^2) & x_{i,3}(a_i^2) \\ \vdots & \vdots & \vdots \\ x_{i,1}(a_i^{n_i}) & x_{i,2}(a_i^{n_i}) & x_{i,3}(a_i^{n_i}) \end{bmatrix}}_{\mathbf{X}_i} \begin{bmatrix} \theta_{i,1} \\ \theta_{i,2} \\ \theta_{i,3} \end{bmatrix} - \underbrace{\begin{bmatrix} y_i(a_i^1) \\ y_i(a_i^2) \\ \vdots \\ y_i(a_i^{n_i}) \end{bmatrix}}_{\mathbf{y}_i} = \underbrace{\begin{bmatrix} e_i^1 \\ e_i^2 \\ \vdots \\ e_i^{n_i} \end{bmatrix}}_{\mathbf{e}_i}$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

- The first-order optimality condition $\nabla_{\theta_i}\left(\tilde{d}_i^2(\theta_i)\right) = 2\sum_{j=1}^{n_i} \nabla_{\theta_i} e_i^j(\theta_i) \max\left\{0,\, e_i^j(\theta_i)\right\} = 0$ can be written as:

$$\mathbf{X}_i^T \left(\max\left\{ \mathbf{0}_{n_i \times 1}, \mathbf{X}_i\theta_i - \mathbf{y}_i \right\} \right) = 0.$$

- In the particular case where $\mathbf{X}_i\theta_i - \mathbf{y}_i = \mathbf{e}_i \geq \mathbf{0}_{n_i \times 1}$ component wise, i.e., when $\mathbf{e}_i = \mathbf{e}_i^+$,

$$\hat{\theta}_i = \left(\mathbf{X}_i^T\mathbf{X}_i\right)^{-1}\mathbf{X}_i^T\mathbf{y}_i$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

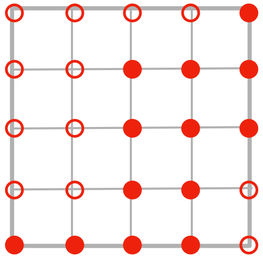
The first-order optimality condition $\nabla_{\theta_i}\left(\tilde{d}_i^{\,2}(\theta_i)\right) = 2\sum_{j=1}^{n_i} \nabla_{\theta_i} e_i^j(\theta_i) \max\left\{0,\, e_i^j(\theta_i)\right\} = 0$ can be written as:

$$\mathbf{X}_i^T \left(\max\left\{ \mathbf{0}_{n_i \times 1}, \mathbf{X}_i \theta_i - \mathbf{y}_i \right\} \right) = 0.$$

In the general case, there will be a subset of samples $J_i(\theta_i) \subset \{1, \ldots, n_1\}$ such that

$$x_i^j \theta_i - y_i^j = e_i^j = e_i^{j,+} \geq 0 \quad \forall j \in J_i(\theta_i)$$

while the remaining samples satisfy the NE condition. Define $\mathbf{X}_i(\theta_i)$ from $\mathbf{X}_i$, keeping only $j \in J_i(\theta_i)$



$\mathscr{A}$

Violate NE conditions

$$\underbrace{\begin{bmatrix} x_{i,1}(a_i^1) & x_{i,2}(a_i^1) & x_{i,3}(a_i^1) \\ \cancel{x_{i,1}(a_i^2)} & \cancel{x_{i,2}(a_i^2)} & \cancel{x_{i,3}(a_i^2)} \\ \vdots & \vdots & \vdots \\ x_{i,1}(a_i^{n_i}) & x_{i,2}(a_i^{n_i}) & x_{i,3}(a_i^{n_i}) \end{bmatrix}}_{\mathbf{X}_i} \qquad e_i^j < 0$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

- The first-order optimality condition $\nabla_{\theta_i}\left(\tilde{d}_i^2(\theta_i)\right) = 2\sum_{j=1}^{n_i} \nabla_{\theta_i} e_i^j(\theta_i) \max\left\{0,\, e_i^j(\theta_i)\right\} = 0$ can be written as:

$$\mathbf{X}_i^T\left(\max\left\{\mathbf{0}_{n_i\times 1}, \mathbf{X}_i\theta_i - \mathbf{y}_i\right\}\right) = 0.$$

- In the general case, there will be a subset of samples $J_i(\theta_i) \subset \{1,\dots,n_1\}$ such that

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while the remaining samples satisfy the NE condition. Define $\mathbf{X}_i(\theta_i)$ from $\mathbf{X}_i$, keeping only $j \in J_i(\theta_i)$

$$\hat{\theta}_i = \left(\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{X}_i(\hat{\theta}_i)\right)^{-1}\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{y}_i(\hat{\theta}_i)$$

INVERSE OPTIMIZATION PROBLEM - SOLUTION

Problem [Minimum NE-Distance Problem].

Given a network $\mathcal{G}^\star$ of N agents, for all agents i find the vectors of preferences $\hat{\theta}_i$ such that

$$\hat{\theta}_i \in \arg\min_{\theta_i \in \Theta} \ \tilde{d}_i^2(\theta_i)$$

The solution

$$\hat{\theta}_i = \left(\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{X}_i(\hat{\theta}_i)\right)^{-1}\mathbf{X}_i^T(\hat{\theta}_i)\mathbf{y}_i(\hat{\theta}_i)$$

still cannot be solved explicitly. However, the (projected) gradient method can be used to reach a solution.

$$\nabla_{\theta_i}\left(\tilde{d}_i^{\,2}(\theta_i)\right) = 2\sum_{j=1}^{n_i} \nabla_{\theta_i} e_i^j(\theta_i) e_i^{j,+}(\theta_i)$$